

COMPETITIVE MATHEMATICS SPECIALIZED CRACKER HANDBOOK

PARTIAL FRACTIONS

The Complete Shortcut, Pattern Recognition & Speed-Solving Master Handbook

Engineered specifically for top-rank aspirants in **JEE Main**, **JEE Advanced**, **TS EAMCET**, and **AP EAMCET**. Designed for rapid 15–45 second problem solving.

JEE Main Edition

JEE Advanced Level

TS / AP EAMCET Speed Drills



3-Step Exam Mastery Blueprint

Step 1: Master the Templates — Memorize the exact algebraic form required for every denominator type instantly (Degree rule: unknowns = degree of denominator).

Step 2: Internalize the 20 Core Shortcuts — Eliminate 80% of textbook simultaneous equations using Cover-Up, Heaviside Derivative, Taylor Shift, and Infinity Limits.

Step 3: Train Instinctive Pattern Recognition — Solve questions in 30–60 seconds using option verification, strategic substitutions ($x = 0, 1, \infty$), and symmetry.

Authored by: Senior Mathematics Faculty (15+ Years Experience Training Top 100 JEE / EAMCET Rankers)
Contains 160+ Curated Problems • 20 High-Speed Shortcuts • 10 Must-Memorize Identities • 30 Common Traps

HOW TO USE THIS HANDBOOK & VISUAL LEGEND

PREPARATION
GUIDE

Exam Weightage & Strategic Impact

While Partial Fractions appears directly as 1–2 independent questions in EAMCET (Algebra section), its real superpower lies as an **essential engine** in **Integral Calculus, Differential Equations, Binomial Theorem for Rational Indices, and Sequences & Series** across JEE Main & Advanced.

Exam	Direct Questions	Calculus & Algebra Applications	Key Priority Focus
TS / AP EAMCET	1 – 2 Direct MCQs	3 – 5 Questions (Integrals / Series)	Lightning Speed (30–45s), Option substitution, Cover-Up
JEE Main	0 – 1 Direct MCQ	4 – 6 Questions (Integration, Telescoping)	Repeated factors, $u = x^2$ substitution, Coefficient sum
JEE Advanced	Rarely standalone	High Frequency in Multi-Concept Calculus	Residues, Taylor shift, Higher-order derivatives, Symmetry

Visual Callout Legend

 **BEST METHOD:** The absolute fastest, highest-speed approach for competitive exams (Target: 15–40s).

 **ALTERNATIVE METHOD:** Reliable backup method (Substitutions / Algebraic comparison).

 **SLOW METHOD:** Standard school textbook method (Full system of linear equations — avoid in exams!).

 **EXAM TRICK:** Smart exam shortcuts, option hacks, and symmetry tricks.

 **COMMON TRAP:** Subtle pitfalls where students lose marks.

 **MUST MEMORIZE:** Core formula to memorize for instant recall.

SECTION 1 — COMPLETE CONCEPT FOUNDATION

THEORY & FOUNDATIONS

1.1 What Are Partial Fractions?

A **rational fraction** is an algebraic expression of the form $R(x) = \frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are polynomials with real coefficients, and $Q(x) \neq 0$.

Partial Fraction Decomposition is the reverse process of adding algebraic fractions. It breaks down a complex rational fraction into a sum of simpler, elementary rational fractions whose denominators are linear or irreducible quadratic polynomials.

Fundamental Theorem of Partial Fractions:

Any proper rational function $\frac{P(x)}{Q(x)}$ (where $\deg(P) < \deg(Q)$) can be uniquely expressed as a finite sum of partial fractions corresponding to the real linear and irreducible quadratic factors of $Q(x)$.

1.2 Proper vs Improper Rational Functions

✓ Proper Rational Function

Degree of Numerator is **strictly less** than Degree of Denominator:

$$\deg(P(x)) < \deg(Q(x))$$

Example: $\frac{2x+3}{x^2+5x+6}$ ($\deg(P) = 1, \deg(Q) = 2$).

👉 **Directly decomposable into standard templates.**

⚠ Improper Rational Function

Degree of Numerator is **greater than or equal to** Degree of Denominator:

$$\deg(P(x)) \geq \deg(Q(x))$$

Example: $\frac{x^3+2}{x^2-1}$ ($\deg(P) = 3 \geq \deg(Q) = 2$).

👉 **MUST divide first before decomposing!**

1.3 Fast Conversion of Improper to Proper Form

By the Polynomial Division Algorithm, any improper fraction can be written as:

$$\frac{P(x)}{Q(x)} = \text{Quotient}(x) + \frac{\text{Remainder}(x)}{Q(x)} \quad \text{where } \deg(R(x)) < \deg(Q(x))$$

SHORTCUT 1: The "Add & Subtract / Term-Matching" Trick (No Long Division!)

When $\deg(P) = \deg(Q)$, match the leading coefficient of $P(x)$ to $Q(x)$ directly in the numerator:

$$\frac{ax^n + \dots}{x^n + \dots} = a + \frac{(P(x) - a \cdot Q(x))}{Q(x)}$$

Basic Example: Fast Conversion

Speed: 10s

Convert $\frac{2x^2+5x+7}{x^2+x-2}$ into proper rational form.

 **Shortcut Solution:** Leading coefficient ratio is $2/1 = 2$.

$$2x^2 + 5x + 7 = 2(x^2 + x - 2) + (3x + 11).$$

Thus:

$$\frac{2x^2 + 5x + 7}{x^2 + x - 2} = 2 + \frac{3x + 11}{(x+2)(x-1)}$$

Competitive Exam Example: Higher Degree Term-Matching

JEE / EAMCET

Find the quotient polynomial when $\frac{x^4+1}{x^2-1}$ is decomposed.

 **Shortcut:** Rewrite numerator in powers of denominator ($x^2 - 1$):

$$x^4 + 1 = (x^2 - 1)(x^2 + 1) + 2.$$

Therefore:

$$\frac{x^4 + 1}{x^2 - 1} = (x^2 + 1) + \frac{2}{x^2 - 1} = x^2 + 1 + \frac{1}{x-1} - \frac{1}{x+1}$$

 **Target Time: 15 seconds (Standard long division takes 60 seconds).**

1.4 Golden Rule for Unknown Constants

The Degree-Count Rule:

The total number of unknown constants ($A, B, C, \dots$) to solve for in the decomposition of a proper rational fraction $\frac{P(x)}{Q(x)}$ is **EXACTLY EQUAL to the degree of the denominator $Q(x)$** .

Checklist: If $Q(x)$ has degree 4, your template MUST contain exactly 4 independent constants (e.g., $A/(x - 1) + B/(x - 2) + (Cx + D)/(x^2 + 1)$ has 4 constants: A, B, C, D).

SECTION 2 — ALL STANDARD FORMS & TEMPLATES

TEMPLATE ARCHITECTURE

Standard Decomposition Templates Reference

In any proper rational fraction $\frac{P(x)}{Q(x)}$, factor the denominator $Q(x)$ completely over real numbers $\mathbb{R}$. Every real polynomial factors into linear factors $(ax + b)$ and irreducible quadratic factors $(ax^2 + bx + c)$ where $b^2 - 4ac < 0$.

Denominator Type	Factor Structure in $Q(x)$	Decomposition Template
Type A: Distinct Linear	$(x - a)(x - b)$	$\frac{A}{x-a} + \frac{B}{x-b}$
Type B: Repeated Linear	$(x - a)^n$	$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \dots + \frac{A_n}{(x-a)^n}$
Type C: Distinct + Repeated Linear	$(x - a)^2(x - b)$	$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \frac{B}{x-b}$
Type D: Irreducible Quadratic	$(x - a)(x^2 + bx + c)$ $(b^2 - 4ac < 0)$	$\frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$
Type E: Repeated Quadratic	$(x^2 + bx + c)^n$	$\sum_{k=1}^n \frac{B_kx+C_k}{(x^2+bx+c)^k}$
Type F: Full Mixture	$(x - a)^2(x - b)(x^2 + c^2)^2$	$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \frac{B}{x-b} + \frac{C_1x+D_1}{x^2+c^2} + \frac{C_2x+D_2}{(x^2+c^2)^2}$

Detailed Breakdown of Each Type

Type A: Distinct Linear Factors

$$\frac{P(x)}{(x-a_1)(x-a_2)\dots(x-a_n)} = \sum_{i=1}^n \frac{A_i}{x-a_i}$$

⚡ 100% solvable by Heaviside Cover-Up in under 15 seconds!

Type B: Repeated Linear Factors

$$\frac{P(x)}{(x-a)^n} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \dots + \frac{A_n}{(x-a)^n}$$

⚡ Solvable instantly via Taylor Shift $x = a + y$.

Type D: Distinct Irreducible Quadratic

Denominator contains $(x^2 + k^2)$ or $(x^2 + bx + c)$.

Numerator **MUST** be linear: $Ax + B$.

⚡ Fast trick: Substitute roots $x = ik$ or use symmetric substitution.

Type E & F: Repeated Quadratics / Mixtures

Each power of quadratic gets its own linear numerator $(B_kx + C_k)$.

Total unknown constants = $\deg(Q(x))$.

Crucial Template Rules & Verification Checklist

CRITICAL TEMPLATE TRAP:

1. Never write $\frac{A}{x^2+1}$ for a quadratic factor — the numerator **MUST be linear** ($Ax + B$) unless the entire expression is a function of x^2 .
2. In repeated factors $(x - a)^3$, you must include **all lower powers**: $\frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{(x-a)^3}$. Missing lower powers is the #1 reason for exam errors!

SECTION 3 — THE 20 FASTEST SHORTCUTS & SPEED METHODS

CORE
ARSENAL

⚡ Shortcut 1 & 2: Heaviside Cover-Up Rule & Derivative Formula

Formula: For distinct linear factor $(x - a_k)$ in $\frac{P(x)}{Q(x)} = \sum \frac{A_k}{x - a_k}$:

$$A_k = \lim_{x \rightarrow a_k} (x - a_k) \frac{P(x)}{Q(x)} = \frac{P(x)}{Q(x) \text{ with } (x - a_k) \text{ removed}} \Big|_{x=a_k} = \frac{P(a_k)}{Q'(a_k)}$$

- **When to Use:** Any distinct linear denominator factors.
- **When NOT to Use:** Irreducible quadratics or non-maximal powers of repeated factors.
- **Exact Procedure:** Cover the factor $(x - a_k)$ with your thumb and substitute $x = a_k$ into the remaining expression.

Worked Example: Cover-Up Method

Target: 10s

Decompose $\frac{3x+1}{(x-1)(x-2)(x-3)}$.

$$A = \frac{3x+1}{(x-2)(x-3)} \Big|_{x=1} = \frac{4}{(-1)(-2)} = 2$$

$$B = \frac{3x+1}{(x-1)(x-3)} \Big|_{x=2} = \frac{7}{(1)(-1)} = -7$$

$$C = \frac{3x+1}{(x-1)(x-2)} \Big|_{x=3} = \frac{10}{(2)(1)} = 5$$

Result: $\frac{2}{x-1} - \frac{7}{x-2} + \frac{5}{x-3}$.

⌚ **Time Saved: 90s (10s vs 100s by system of 3 equations).**

⚡ Shortcut 3 & 4: Smart Strategic Value Substitution

Concept: Instead of expanding polynomials into equations, substitute strategic values: $x = 0, 1, -1$, or evaluate limits at $x \rightarrow \infty$.

- **When to Use:** To find remaining constants after Heaviside Cover-up has yielded the easy ones.
- **Exact Procedure:** 1. Find maximal coefficients via Cover-Up. 2. Plug $x = 0$ (simplifies all x terms to constants). 3. Compare leading coefficients or plug $x = 1$.

Worked Example: Strategic Evaluation

Target: 20s

Find A, B, C in $\frac{x^2+1}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$.

1. Cover-up for B (at $x = 1$): $B = \frac{1^2+1}{1+2} = \frac{2}{3}$.
2. Cover-up for C (at $x = -2$): $C = \frac{(-2)^2+1}{(-2-1)^2} = \frac{5}{9}$.
3. Smart Value at $x = 0$: $\frac{1}{(1)(2)} = \frac{A}{-1} + \frac{2/3}{1} + \frac{5/9}{2} \Rightarrow \frac{1}{2} = -A + \frac{2}{3} + \frac{5}{18}$
 $A = \frac{12+5-9}{18} = \frac{8}{18} = \frac{4}{9}$.

🕒 **Time Saved: 75s (20s vs 95s).**

⚡ Shortcut 5 & 6: Highest-Degree & Constant Term Comparison

Key Idea: In $P(x) = A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b)$:

- Coefficient of x^2 gives: $\text{Coeff}(x^2) = A + B + C$.
- Constant term (at $x = 0$) gives: $P(0) = A(bc) + B(ac) + C(ab)$.
- **When to Use:** When a question asks for the sum of all coefficients $A + B + C$ or when determining the highest-order coefficient instantly.
- **Time Saved:** 60–90 seconds (Instant 5-second inspection).

⚡ Shortcut 7 & 8: Derivative / Differential Operator & Residue Formula

General Differential Formula for Repeated Factor $(x - a)^n$:

$$A_{n-k} = \frac{1}{k!} \frac{d^k}{dx^k} \left[(x-a)^n \frac{P(x)}{Q(x)} \right] \bigg|_{x=a} \quad \text{for } k = 0, 1, 2, \dots, n-1$$

• Specifically for $k = 1$ (Linear term under $(x-a)^2$):

$$A_1 = \frac{d}{dx} \left[(x-a)^2 \frac{P(x)}{Q(x)} \right] \bigg|_{x=a}$$

Worked Example: Derivative Extraction

Target: 25s

Find the coefficient of $\frac{1}{x-1}$ in $\frac{x+3}{(x-1)^2(x+1)}$.

$$\text{Let } f(x) = (x-1)^2 \cdot \frac{x+3}{(x-1)^2(x+1)} = \frac{x+3}{x+1} = 1 + \frac{2}{x+1}.$$

$$\text{Differentiating with respect to } x: f'(x) = -\frac{2}{(x+1)^2}.$$

$$\text{Evaluate at } x = 1: A_1 = f'(1) = -\frac{2}{(1+1)^2} = -\frac{2}{4} = -\frac{1}{2}.$$

🕒 **Time Saved: 80s (Done in 1 line without solving for the other fractions!).**

⚡ Shortcut 9 & 10: Limit at Infinity & Intelligent Root Elimination

The Infinite Limit Shortcut:

$$\lim_{x \rightarrow \infty} x \cdot \frac{P(x)}{Q(x)} = \sum_{i=1}^n A_i$$

(when $\deg(Q) = \deg(P) + 1$, the sum of all linear partial fraction coefficients equals $\frac{\text{Leading coeff of } P}{\text{Leading coeff of } Q}$).

• **Direct Application:** If $\frac{2x+5}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, then multiplying by x and letting $x \rightarrow \infty$ instantly yields:

$$A + B + C = 0 \quad (\text{since } \deg(P) < \deg(Q) - 1)$$

If numerator was $2x^2 + 5$, then $A + B + C = 2$.

⚡ Shortcut 11 & 12: Even Power Symmetry Trick ($u = x^2$)

The Even-Degree Rule: If ALL powers of x in $\frac{P(x)}{Q(x)}$ are **EVEN**, substitute $u = x^2$. Decompose into **linear** partial fractions in u , then substitute back $u = x^2$.

• **Danger Check:** This is an *algebraic decomposition shortcut*. If integrating, you cannot replace dx with du directly without proper chain rule, but you DO integrate the resulting quadratic fractions $\int \frac{A}{x^2+a^2} dx$.

Worked Example: Even Powers Shortcut

Target: 15s

Decompose $\frac{x^2+2}{(x^2+1)(x^2+4)}$.

Put $u = x^2$: $\frac{u+2}{(u+1)(u+4)} = \frac{A}{u+1} + \frac{B}{u+4}$.

By Cover-Up on u :

$$A = \left. \frac{u+2}{u+4} \right|_{u=-1} = \frac{1}{3}, \quad B = \left. \frac{u+2}{u+1} \right|_{u=-4} = \frac{-2}{-3} = \frac{2}{3}.$$

Replace $u = x^2$:

$$\frac{x^2+2}{(x^2+1)(x^2+4)} = \frac{1/3}{x^2+1} + \frac{2/3}{x^2+4}$$

⌚ **Time Saved: 100s (Avoiding 4-variable system $Ax + B, Cx + D$).**

⚡ Shortcut 13 & 14: Factoring Shortcuts & Pre-Cancellation

Must-Know Competitive Exam Factorizations:

1. $x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$
2. $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$ (Sophie Germain Identity)
3. $x^3 \pm 1 = (x \pm 1)(x^2 \mp x + 1)$
4. $x^4 - 1 = (x - 1)(x + 1)(x^2 + 1)$

Pre-Cancellation Rule: Always check if $P(x)$ and $Q(x)$ share a common factor before decomposing. E.g., $\frac{x^2-1}{x^3-1} = \frac{(x-1)(x+1)}{(x-1)(x^2+x+1)} = \frac{x+1}{x^2+x+1}$. Decomposing before cancelling creates ghost singularities!

⚡ Shortcut 15 & 16: Complex Roots for Quadratic Factors

Concept: For irreducible quadratic $x^2 + a^2$, its roots are $x = \pm ia$. Treating $x = ia$ in the cover-up method and equating real and imaginary parts extracts B and C in $\frac{Bx+C}{x^2+a^2}$ in ONE STEP!

Worked Example: Complex Root Extraction

Target: 30s

Find B, C in $\frac{x}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$.

1. Cover-up for A at $x = 1$: $A = \frac{1}{1^2+1} = \frac{1}{2}$.

2. Plug $x = i$ (the root of $x^2 + 1 = 0$):

$$\frac{x}{x-1} \Big|_{x=i} = Bi + C \Rightarrow \frac{i}{i-1} = \frac{i(i+1)}{(i-1)(i+1)} = \frac{-1+i}{-2} = \frac{1}{2} - \frac{1}{2}i.$$

Equating real and imaginary parts: $C = \frac{1}{2}, B = -\frac{1}{2}$.

⌚ **Time Saved: 60s (Completely avoids simultaneous equations!).**

⚡ Shortcut 17 & 18: Taylor Shift Substitution ($x = a + y$)

The "Shift to Origin" Method for Repeated Factors:

To decompose $\frac{P(x)}{(x-a)^n}$ or $\frac{P(x)}{(x-a)^n(x-b)}$, substitute $x = a + y$. Expand the numerator in powers of y , divide term-by-term, then replace $y = x - a$.

Worked Example: Taylor Shift Trick

Target: 20s

Decompose $\frac{x^2+3x+5}{(x-2)^3}$.

Let $x - 2 = y \Rightarrow x = y + 2$.

Numerator: $(y+2)^2 + 3(y+2) + 5 = y^2 + 4y + 4 + 3y + 6 + 5 = y^2 + 7y + 15$.

Divide by y^3 :

$$\frac{y^2 + 7y + 15}{y^3} = \frac{1}{y} + \frac{7}{y^2} + \frac{15}{y^3} = \frac{1}{x-2} + \frac{7}{(x-2)^2} + \frac{15}{(x-2)^3}$$

⌚ **Time Saved: 70s (Solvable purely mentally in 15 seconds!).**

⚡ Shortcut 19 & 20: Finding Combinations ($A + B, AB$) & Partial Decomposition

Target-Oriented Extraction:

When an exam asks for $A + B$, $A - B + C$, or $A \cdot B$, **NEVER** solve for individual constants separately!

- For $A + B + C$: Take $\lim_{x \rightarrow \infty} x \cdot f(x)$.
- For $A - B + C$: Plug $x = -1$ or an appropriate evaluation point.
- For isolated constant A_k : Apply Heaviside Cover-Up directly on $(x - a_k)$ and immediately ignore the rest of the expression.

SECTION 4 — HIGH-YIELD IDENTITIES (MUST MEMORIZE BOX)

INSTANT
RECALL

Top 10 High-Frequency Partial Fraction Identities

Memorizing these 10 exact algebraic identities eliminates calculation completely during JEE Main & EAMCET.

1. Two Linear Factors (Constant Numerator):

$$\frac{1}{(x-a)(x-b)} = \frac{1}{b-a} \left[\frac{1}{x-b} - \frac{1}{x-a} \right] = \frac{1}{a-b} \left[\frac{1}{x-a} - \frac{1}{x-b} \right]$$

Mnemonic: $\frac{1}{\text{Difference}} \left(\frac{1}{\text{Smaller root factor}} - \frac{1}{\text{Larger root factor}} \right)$

2. Two Linear Factors (Linear Numerator):

$$\frac{x}{(x-a)(x-b)} = \frac{1}{a-b} \left[\frac{a}{x-a} - \frac{b}{x-b} \right]$$

Mnemonic: Each numerator gets its own root! Root a on top of $(x-a)$, root b on top of $(x-b)$.

3. Standard Difference of Squares:

$$\frac{1}{x^2 - a^2} = \frac{1}{(x-a)(x+a)} = \frac{1}{2a} \left[\frac{1}{x-a} - \frac{1}{x+a} \right]$$

$$\frac{x}{x^2 - a^2} = \frac{1}{2} \left[\frac{1}{x-a} + \frac{1}{x+a} \right]$$

4. Three Linear Factors (Symmetric Form):

$$\frac{1}{(x-a)(x-b)(x-c)} = \frac{1}{(a-b)(a-c)(x-a)} + \frac{1}{(b-a)(b-c)(x-b)} + \frac{1}{(c-a)(c-b)(x-c)}$$

Mnemonic: Cyclic permutation of (a, b, c) . Notice sum of coefficients is 0!

5. Consecutive Integer Linear Factors (Telescoping Series Form):

$$\frac{1}{x(x+1)(x+2)} = \frac{1}{2} \left[\frac{1}{x} - \frac{2}{x+1} + \frac{1}{x+2} \right]$$

Coefficients follow Pascal triangle row signs: $+1, -2, +1$, scaled by $\frac{1}{2!}$.

6. Four Consecutive Factors:

$$\frac{1}{x(x+1)(x+2)(x+3)} = \frac{1}{6} \left[\frac{1}{x} - \frac{3}{x+1} + \frac{3}{x+2} - \frac{1}{x+3} \right]$$

Binomial coefficients: $+1, -3, +3, -1$ scaled by $\frac{1}{3!} = \frac{1}{6}$.

7. Repeated Linear Factor with Distinct Factor:

$$\frac{1}{(x-a)^2(x-b)} = \frac{1}{(a-b)^2} \left[\frac{1}{x-b} - \frac{1}{x-a} \right] - \frac{1}{(a-b)(x-a)^2}$$

8. Two Pure Quadratic Factors (Even Powers):

$$\frac{1}{(x^2+a^2)(x^2+b^2)} = \frac{1}{b^2-a^2} \left[\frac{1}{x^2+a^2} - \frac{1}{x^2+b^2} \right]$$

$$\frac{x^2}{(x^2+a^2)(x^2+b^2)} = \frac{1}{b^2-a^2} \left[\frac{b^2}{x^2+b^2} - \frac{a^2}{x^2+a^2} \right]$$

9. Quadratic Factor Form:

$$\frac{1}{x(x^2 + a^2)} = \frac{1}{a^2} \left[\frac{1}{x} - \frac{x}{x^2 + a^2} \right]$$

$$\frac{1}{x(x^n + 1)} = \frac{1}{x} - \frac{x^{n-1}}{x^n + 1}$$

\quad (\text{Integration Standard Form})

10. General Polynomial Cover-Up (Heaviside Form):

$$\frac{P(x)}{(x - a_1)(x - a_2) \dots (x - a_n)} = \sum_{k=1}^n \frac{P(a_k)}{\prod_{j \neq k} (a_k - a_j)} \cdot \frac{1}{x - a_k}$$

SECTION 5 — SPECIAL SHORTCUTS & ADVANCED TECHNIQUES

ADVANCED
TOOLKIT

5.1 The Derivative Form: $A_k = \frac{P(a_k)}{Q'(a_k)}$

The Differential Residue Theorem:

If $Q(x)$ has non-repeated roots $a_1, a_2, \dots, a_n$, then the partial fraction coefficient of $\frac{1}{x-a_k}$ is:

$$A_k = \frac{P(a_k)}{Q'(a_k)}$$

Why this is a superpower: You do NOT need to factor the rest of the denominator! If you know just ONE root a_k , differentiate $Q(x)$ and plug $x = a_k$.

Example: Fast Extraction with $Q'(x)$

Speed: 15s

Find the coefficient of $\frac{1}{x-2}$ in the decomposition of $\frac{x^3+5}{x^5-32}$.

Here $P(x) = x^3 + 5$ and $Q(x) = x^5 - 32$ (root is $x = 2$).

$$Q'(x) = 5x^4.$$

Therefore:

$$A = \frac{P(2)}{Q'(2)} = \frac{2^3 + 5}{5(2^4)} = \frac{8 + 5}{5(16)} = \frac{13}{80}$$

 **Target Time: 15 seconds (Standard polynomial division/factoring would take 2+ minutes!).**

5.2 Rapid Sums of Coefficients Without Individual Solving

Theorem on Sum of Coefficients:

Let $\frac{P(x)}{(x-a_1)(x-a_2)\dots(x-a_n)} = \sum_{k=1}^n \frac{A_k}{x-a_k}$.

- If $\deg(P) < n - 1$, then

$$\sum_{k=1}^n A_k = 0$$

- If $\deg(P) = n - 1$, then

$$\sum_{k=1}^n A_k = \text{Leading Coefficient of } P(x)$$

- The weighted sum $\sum_{k=1}^n A_k a_k = \text{Coefficient of } x^{n-2} \text{ in } P(x) - (\sum a_i) \cdot (\dots)$

5.3 Palindromic & Symmetric Denominators ($x + \frac{1}{x} = t$)

When rational expressions have symmetric coefficients like $x^4 + ax^3 + bx^2 + ax + 1$, divide numerator and denominator by x^2 and substitute $t = x + \frac{1}{x}$ where $x^2 + \frac{1}{x^2} = t^2 - 2$.

Example: Symmetrical Rational Function

JEE Advanced

Decompose $\frac{x^2-1}{x^4+1}$.

Divide numerator and denominator by x^2 :

$$\frac{1 - \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} = \frac{d(x + 1/x)}{(x + 1/x)^2 - 2}$$

With $t = x + 1/x$, this is $\frac{1}{t^2-2} = \frac{1}{2\sqrt{2}} \left[\frac{1}{t-\sqrt{2}} - \frac{1}{t+\sqrt{2}} \right]$.

Converting back to x :

$$\frac{x^2-1}{x^4+1} = \frac{1}{2\sqrt{2}} \left[\frac{x}{x^2 - \sqrt{2}x + 1} - \frac{x}{x^2 + \sqrt{2}x + 1} \right]$$

5.4 Telescoping Partial Fractions & Series Summation

General Product Partial Fraction Identity for Series:

$$\frac{1}{r(r+1)(r+2)\dots(r+k)} = \frac{1}{k} \left[\frac{1}{r(r+1)\dots(r+k-1)} - \frac{1}{(r+1)(r+2)\dots(r+k)} \right]$$

This telescoping difference cancels all intermediate terms when summing from $r = 1$ to n :

$$\sum_{r=1}^n \frac{1}{r(r+1)(r+2)} = \frac{1}{2} \sum_{r=1}^n \left[\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)} \right] = \frac{1}{2} \left[\frac{1}{1 \cdot 2} - \frac{1}{(n+1)(n+2)} \right]$$

SECTION 6 — JEE MAIN SHORTCUTS & 25 REPRESENTATIVE QUESTIONS

JEE MAIN
SPEED LAB

Q1. Distinct Linear Factors with Linear Numerator

★ **MUST KNOW**

🔥 **HIGH FREQUENCY**

⚡ **FASTEST METHOD**

🕒 30s

If $\frac{3x-1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$, then the value of $A^2 + B^2$ is:

A) 74

B) 89

C) 65

D) 50

⚡ **15-Second Shortcut:** Heaviside Cover-Up: $A = \frac{3x-1}{x-3} \Big|_{x=2} = \frac{5}{-1} = -5$, $B = \frac{3x-1}{x-2} \Big|_{x=3} = \frac{8}{1} = 8$.

Detailed Solution:

1. Evaluate $A = -5$ and $B = 8$.

2. Compute $A^2 + B^2 = (-5)^2 + 8^2 = 25 + 64 = 89$.

⚠ **Common Mistake:** Solving simultaneous equations $A + B = 3$, $-3A - 2B = -1$ which wastes 45 seconds.

🏆 **BEST: Cover-Up (10s)** | 🐢 **SLOW: Equating coefficients (55s)**

Q2. Three Linear Factors — Single Required Constant

🔥 **HIGH FREQUENCY**

⚡ **FASTEST METHOD**

🕒 25s

If $\frac{2x^2+5x-3}{(x-1)(x-2)(x+3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+3}$, find the value of B .

A) 3

B) 15/5

C) 3

D) 15

⚡ **15-Second Shortcut:** Targeted Cover-Up only at $x = 2$: $B = \frac{2x^2+5x-3}{(x-1)(x+3)} \Big|_{x=2} = \frac{2(4)+10-3}{(1)(5)} = \frac{15}{5} = 3$.

Detailed Solution:

Do NOT solve for A or C . Directly plug $x = 2$ into the covered expression to obtain $B = 3$.

⚠ **Common Mistake:** Computing all three coefficients A, B, C when only B was requested.

🏆 **BEST: Isolated Cover-Up (12s)** | 🐢 **SLOW: Full 3x3 linear system (90s)**

Q3. Sum of All Partial Fraction Coefficients★ **MUST KNOW**⚡ **FASTEST METHOD**🕒 **15s**

If $\frac{5x^2-3x+2}{(x-1)(x-2)(x-4)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-4}$, then $A + B + C$ equals:

A) 0

B) 2

C) 5

D) -3

⚡ **15-Second Shortcut:** Multiply by x and take $x \rightarrow \infty$: $A + B + C = \lim_{x \rightarrow \infty} \frac{x(5x^2-3x+2)}{x^3-7x^2+\dots} = 5$.

Detailed Solution:

Leading degree of numerator is 2, leading degree of denominator is 3. The limit as $x \rightarrow \infty$ is the ratio of leading coefficients: $5/1 = 5$.

⚠ **Common Mistake:** Calculating A, B, C individually and adding them up (takes 80s vs 5s).

🏆 **BEST: Limit at Infinity (5s)** | 🐢 **SLOW: Individual Cover-Up & Sum (75s)**

Q4. Pure Even Powers Substitution ($u = x^2$)🔥 **HIGH FREQUENCY**★ **MUST KNOW**🕒 **35s**

If $\frac{x^2+5}{(x^2+2)(x^2+3)} = \frac{A}{x^2+2} + \frac{B}{x^2+3}$, then $A - B$ equals:

A) 1

B) 3

C) 5

D) -1

⚡ **15-Second Shortcut:** Substitute $u = x^2$: $\frac{u+5}{(u+2)(u+3)} = \frac{A}{u+2} + \frac{B}{u+3}$. Cover-up: $A = \frac{-2+5}{-2+3} = 3$, $B = \frac{-3+5}{-3+2} = -2$.

Detailed Solution:

1. $A = 3, B = -2$.

2. $A - B = 3 - (-2) = 5$.

⚠ **Common Mistake:** Assuming $\frac{Ax+B}{x^2+2} + \frac{Cx+D}{x^2+3}$ and solving 4 linear equations.

🏆 **BEST: $u = x^2$ reduction (20s)** | 🐢 **SLOW: 4-variable system (110s)**

Q5. Repeated Linear Factor — Highest Power Term**🔥 HIGH FREQUENCY****🕒 25s**

In the partial fractions of $\frac{x^2-2x+4}{(x-1)^3}$ as $\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}$, the value of $A + B + C$ is:

A) 1

B) 3

C) 4

D) 0

⚡ 15-Second Shortcut: Taylor Shift $x = 1 + y$: $(1+y)^2 - 2(1+y) + 4 = 1 + 2y + y^2 - 2 - 2y + 4 = y^2 + 3$.

Divide by y^3 : $\frac{1}{y} + \frac{0}{y^2} + \frac{3}{y^3}$. Thus $A = 1, B = 0, C = 3$.

Detailed Solution:

1. $A = 1, B = 0, C = 3$.

2. $A + B + C = 1 + 0 + 3 = 4$.

⚠️ Common Mistake: Expanding $A(x-1)^2 + B(x-1) + C$ and comparing coefficients.

🏆 BEST: Taylor Shift $x = 1 + y$ (15s) | **🐢 SLOW:** Polynomial expansion (60s)

Q6. Irreducible Quadratic Factor via Complex Roots**★ MUST KNOW****⚡ FASTEST METHOD****🕒 40s**

If $\frac{1}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$, then $B + C$ is:

A) -2/5

B) -1/20

C) 1/5

D) -1/5

⚡ 15-Second Shortcut: 1. $A = \frac{1}{x^2+4} \Big|_{x=1} = \frac{1}{5}$.

2. Plug $x = 2i$: $\frac{1}{2i-1} = 2iB + C \Rightarrow \frac{-1-2i}{5} = C + 2iB \Rightarrow C = -\frac{1}{5}, B = -\frac{1}{5}$.

3. $B + C = -\frac{1}{5} - \frac{1}{5} = -\frac{2}{5}$.

Detailed Solution:

1. $A = 1/5$.

2. At $x \rightarrow \infty$: $A + B = 0 \Rightarrow B = -1/5$.

3. At $x = 0$: $-\frac{1}{4} = -A + \frac{C}{4} \Rightarrow C = -1/5$.

4. $B + C = -2/5$.

⚠️ Common Mistake: Solving 3 equations by multiplying out full polynomial.

🏆 BEST: Complex root or $x = 0, \infty$ (20s) | **🐢 SLOW:** 3x3 matrix (70s)

Q7. Improper Fraction Decomposition **HIGH FREQUENCY** **35s**

If $\frac{x^3+1}{x^2-3x+2} = (x+a) + \frac{b}{x-1} + \frac{c}{x-2}$, find $a+b+c$.

A) 7

B) 10

C) 12

D) 14

15-Second Shortcut: 1. Polynomial division: $x^3 + 1 = (x+3)(x^2 - 3x + 2) + (7x - 5) \Rightarrow a = 3$.

2. Cover-up on remainder: $b = \frac{7x-5}{x-2} \Big|_{x=1} = \frac{2}{-1} = -2$, $c = \frac{7x-5}{x-1} \Big|_{x=2} = \frac{9}{1} = 9$.

Detailed Solution:

1. $a = 3, b = -2, c = 9$.

2. $a + b + c = 3 + (-2) + 9 = 10$.

Common Mistake: Decomposing without dividing the improper fraction first.

BEST: Synthetic division + Cover-up (30s) | **SLOW:** Undetermined coefficients (90s)

Q8. Integration Application of Partial Fractions **MUST KNOW** **HIGH FREQUENCY** **40s**

If $\int \frac{2x+3}{(x-1)(x+2)} dx = A \ln|x-1| + B \ln|x+2| + K$, then $A - B$ equals:

A) 1

B) 2/3

C) 4/3

D) 1/3

15-Second Shortcut: By Cover-Up: $A = \frac{2x+3}{x+2} \Big|_{x=1} = \frac{5}{3}$, $B = \frac{2x+3}{x-1} \Big|_{x=-2} = \frac{-1}{-3} = \frac{1}{3}$.

$A - B = \frac{5}{3} - \frac{1}{3} = \frac{4}{3}$.

Detailed Solution:

Coefficients of logarithmic terms in integration of distinct linear factors are simply the partial fraction constants!

Common Mistake: Doing indefinite integration steps before decomposing.

BEST: Instant Cover-Up (15s) | **SLOW:** Standard integration steps (60s)

Q9. Factoring Trick: Denominator with $x^4 - 16$ **HIGH FREQUENCY** **45s**

If $\frac{x^2}{(x^2-4)(x^2+4)} = \frac{A}{x-2} + \frac{B}{x+2} + \frac{Cx+D}{x^2+4}$, find $A + B$.

A) 0

B) 1/4

C) 1/8

D) 1/2

15-Second Shortcut: The function is EVEN in x . For any even function decomposed into linear terms, $B = -A \Rightarrow A + B = 0$.

Detailed Solution:

By even symmetry: $f(-x) = f(x) \Rightarrow \frac{A}{-x-2} + \frac{B}{-x+2} = \frac{-A}{x+2} - \frac{B}{x-2}$. Comparing terms gives $A = -B \Rightarrow A + B = 0$.

Common Mistake: Evaluating all 4 constants individually.

BEST: Parity / Even symmetry (5s) | **SLOW:** 4-term Cover-up (60s)

Q10. Repeated Quadratic Denominator Term **MUST KNOW** **45s**

In $\frac{x^3+2x}{(x^2+1)^2} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2}$, find $A + B + C + D$.

A) 1

B) 2

C) 3

D) 0

15-Second Shortcut: Let $t = x^2 + 1 \Rightarrow x^2 = t - 1$.

Numerator $x^3 + 2x = x(x^2 + 2) = x((t - 1) + 2) = x(t + 1) = xt + x$.

Divide by t^2 : $\frac{xt+x}{t^2} = \frac{x}{t} + \frac{x}{t^2} = \frac{x}{x^2+1} + \frac{x}{(x^2+1)^2}$.

Detailed Solution:

Matching coefficients: $A = 1, B = 0, C = 1, D = 0$.

$A + B + C + D = 1 + 0 + 1 + 0 = 2$.

Common Mistake: Multiplying out $(Ax + B)(x^2 + 1) + (Cx + D)$ and solving 4 linear equations.

BEST: Algebraic substitution $t = x^2 + 1$ (20s) | **SLOW:** 4-variable expansion (80s)

Q11. Derivative Shortcut for Linear Factor in Repeated Setting

⚡ FASTEST METHOD

★ MUST KNOW

🕒 40s

If $\frac{2x+1}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$, find A .

A) 1/9

B) -1/9

C) 1/3

D) -1/3

⚡ **15-Second Shortcut:** By limit at $x \rightarrow \infty$: $A + C = 0 \Rightarrow A = -C$. Since $C = \left. \frac{2x+1}{(x-1)^2} \right|_{x=-2} = \frac{-3}{9} = -\frac{1}{3}$, we have $A = \frac{1}{3}$.

Detailed Solution:

Derivative method also confirms: $A = \left. \frac{d}{dx} \left[\frac{2x+1}{x+2} \right] \right|_{x=1} = \left. \frac{2(x+2) - (2x+1)}{(x+2)^2} \right|_{x=1} = \frac{3}{9} = \frac{1}{3}$.

⚠ **Common Mistake:** Differentiating incorrectly or expanding all brackets.

🏆 **BEST:** $A = -C$ from $x \rightarrow \infty$ (10s) | 🐢 **SLOW:** Bracket expansion (60s)

Q12. Symmetric Product of Linear Factors

🔥 HIGH FREQUENCY

🕒 30s

If $\frac{1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, then $A : B : C$ is:

A) 1 : -2 : 1

B) 1 : -1 : 1

C) 1 : 2 : 1

D) -1 : 2 : -1

⚡ **15-Second Shortcut:** By Must-Memorize Identity 5 (equispaced roots): Coefficients are proportional to binomial alternating signs +1, -2, +1.
 $A = 1/2, B = -1, C = 1/2 \Rightarrow 1 : -2 : 1$.

Detailed Solution:

Cover-up: $A = \frac{1}{(-1)(-2)} = \frac{1}{2}$, $B = \frac{1}{(1)(-1)} = -1$, $C = \frac{1}{(2)(1)} = \frac{1}{2}$. Ratio: $1/2 : -1 : 1/2 = 1 : -2 : 1$.

⚠ **Common Mistake:** Solving algebraically when binomial pattern gives it in 5 seconds.

🏆 **BEST:** Binomial pattern recall (5s) | 🐢 **SLOW:** Cover-up manual calculation (25s)

Q13. Binomial Series Application of Partial Fractions★ **MUST KNOW**

🕒 50s

Coefficient of x^n in the expansion of $\frac{3x+1}{(1-x)(1-2x)}$ in ascending powers of x is:

A) $7 \cdot 2^n - 4$

B) $7 \cdot 2^n + 4$

C) $4 \cdot 2^n - 7$

D) $2^{n+1} - 1$

⚡ **15-Second Shortcut:** Cover-up: $\frac{3x+1}{(1-x)(1-2x)} = \frac{A}{1-x} + \frac{B}{1-2x}$.

$$A = \left. \frac{3x+1}{1-2x} \right|_{x=1} = \frac{4}{-1} = -4, B = \left. \frac{3x+1}{1-x} \right|_{x=1/2} = \frac{5/2}{1/2} = 7.$$

Coefficient of x^n is $-4(1)^n + 7(2^n) = 7 \cdot 2^n - 4$.

Detailed Solution:

Expand each partial fraction as geometric series: $A(1-x)^{-1} + B(1-2x)^{-1} = A \sum x^n + B \sum (2x)^n$.

⚠ **Common Mistake:** Multiplying polynomials instead of decomposing into geometric series.

🏆 **BEST:** Cover-Up + Geometric Series (25s) | 🐢 **SLOW:** Long division (90s)

Q14. Sophie Germain Denominator🔥 **HIGH FREQUENCY**⚡ **FASTEST METHOD****Factorization**

🕒 50s

The partial fraction decomposition of $\frac{4x}{x^4+4}$ contains which of the following terms?

A) $\frac{1}{x^2-2x+2} - \frac{1}{x^2+2x+2}$

B) $\frac{1}{x^2-2x+2} + \frac{1}{x^2+2x+2}$

C) $\frac{x}{x^2-2x+2}$

D) $\frac{2}{x^2+4}$

⚡ **15-Second Shortcut:** Denominator: $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$.

Difference between factors is $(x^2 + 2x + 2) - (x^2 - 2x + 2) = 4x$.

$$\text{Thus } \frac{4x}{(x^2-2x+2)(x^2+2x+2)} = \frac{1}{x^2-2x+2} - \frac{1}{x^2+2x+2}.$$

Detailed Solution:

Notice the numerator is EXACTLY the difference of the two quadratic factors! Decomposition is immediate.

⚠ **Common Mistake:** Setting up $\frac{Ax+B}{x^2-2x+2} + \frac{Cx+D}{x^2+2x+2}$ and solving 4 equations.

🏆 **BEST:** Factor difference inspection (10s) | 🐢 **SLOW:** 4x4 linear system (100s)

Q15. Difference of Factors in Numerator★ **MUST KNOW**⚡ **FASTEST METHOD**

🕒 25s

If $\frac{1}{x(x+1)} + \frac{1}{(x+1)(x+2)} + \frac{1}{(x+2)(x+3)} = \frac{A}{x} + \frac{B}{x+3}$, then $A + B$ is:

A) 0

B) 1

C) 3

D) -3

⚡ **15-Second Shortcut:** Telescoping series: $(\frac{1}{x} - \frac{1}{x+1}) + (\frac{1}{x+1} - \frac{1}{x+2}) + (\frac{1}{x+2} - \frac{1}{x+3}) = \frac{1}{x} - \frac{1}{x+3}$.
Here $A = 1, B = -1 \Rightarrow A + B = 0$.

Detailed Solution:

Internal terms cancel completely, leaving $\frac{1}{x} - \frac{1}{x+3}$.

⚠ **Common Mistake:** Taking LCM of all three denominators and then decomposing.

🏆 **BEST:** Telescoping identity (10s) | 🐢 **SLOW:** Full LCM addition (75s)

Q16. Quadratic Factor with Linear Numerator🔥 **HIGH FREQUENCY**

🕒 35s

If $\frac{x^2-x+2}{(x^2+1)(x-1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$, then $A + B + C$ is:

A) 0

B) 1

C) 2

D) -1

⚡ **15-Second Shortcut:** 1. Cover-up for A at $x = 1$: $A = \frac{1-1+2}{1+1} = 1$.
2. Leading degree limit $x \rightarrow \infty$: $A + B = 1 \Rightarrow 1 + B = 1 \Rightarrow B = 0$.
3. Plug $x = 0$: $\frac{2}{(1)(-1)} = \frac{A}{-1} + \frac{C}{1} \Rightarrow -2 = -1 + C \Rightarrow C = -1$.
4. $A + B + C = 1 + 0 + (-1) = 0$.

Detailed Solution:

1. $A = 1$. 2. $B = 0$. 3. $C = -1$. Sum $A + B + C = 0$.

⚠ **Common Mistake:** Expanding the full numerator.

🏆 **BEST:** Cover-up + Limit + $x = 0$ (20s) | 🐢 **SLOW:** Full expansion (65s)

Q17. Evaluating at $x = 0$ for Constant Product★ **MUST KNOW**

🕒 30s

If $\frac{1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, then the value of $6A + 3B + 2C$ is:

A) 1

B) -1

C) 0

D) 6

⚡ **15-Second Shortcut:** Multiply the equation by 6 and plug $x = 0$:

$$\frac{6}{(-1)(-2)(-3)} = \frac{6A}{-1} + \frac{6B}{-2} + \frac{6C}{-3} \Rightarrow -1 = -(6A + 3B + 2C) \Rightarrow 6A + 3B + 2C = 1.$$

Detailed Solution:

Setting $x = 0$ directly matches the requested linear combination without finding A, B, C !

⚠ **Common Mistake:** Computing $A = 1/2, B = -1, C = 1/2$ and then substituting (waste of time).

🏆 **BEST:** Smart substitution $x = 0$ (10s) | 🐢 **SLOW:** Individual calculation (40s)

Q18. Three Factors with Quadratic Numerator🔥 **HIGH FREQUENCY**

🕒 40s

If $\frac{x^2}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, then $A + B + C$ is:

A) 1

B) 0

C) 2

D) 3

⚡ **15-Second Shortcut:** Multiply by x and take $x \rightarrow \infty$:

$$A + B + C = \lim_{x \rightarrow \infty} \frac{x \cdot x^2}{(x-1)(x-2)(x-3)} = \lim_{x \rightarrow \infty} \frac{x^3}{x^3 + \dots} = 1.$$

Detailed Solution:

The sum of coefficients of proper rational fractions with $\deg(P) = \deg(Q) - 1$ is always the ratio of leading coefficients ($1/1 = 1$).

⚠ **Common Mistake:** Calculating $A = 1/2, B = -4, C = 9/2$ and then adding $1/2 - 4 + 9/2 = 1$.

🏆 **BEST:** $x \rightarrow \infty$ limit (4s) | 🐢 **SLOW:** Cover-up and addition (45s)

Q19. Cancellation of Linear Factor Before Decomposition★ **MUST KNOW**

🕒 25s

Decompose $\frac{x^2-3x+2}{x^3-4x^2+5x-2}$.

A) $\frac{1}{x-1}$

B) $\frac{1}{x-2}$

C) $\frac{1}{x-1} + \frac{1}{x-2}$

D) $\frac{1}{(x-1)^2}$

⚡ **15-Second Shortcut:** Factor numerator: $(x-1)(x-2)$.Factor denominator: $(x-1)^2(x-2)$.Cancel $(x-1)(x-2)$: $\frac{(x-1)(x-2)}{(x-1)^2(x-2)} = \frac{1}{x-1}$.**Detailed Solution:**

Always factor and cancel common terms first before applying partial fractions!

⚠ **Common Mistake:** Setting up a 3-term partial fraction $\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2}$.🏆 **BEST:** Pre-cancellation (10s) | 🐢 **SLOW:** Full decomposition + cancellation (70s)**Q20. Partial Fractions in Binomial Expansion**🔥 **HIGH FREQUENCY**

🕒 45s

If $\frac{1}{(1-x)(1-2x)(1-3x)} = \frac{A}{1-x} + \frac{B}{1-2x} + \frac{C}{1-3x}$, then $A+B+C$ is:

A) 0

B) 1

C) 3

D) 6

⚡ **15-Second Shortcut:** Substitute $x=0$: $\frac{1}{(1)(1)(1)} = \frac{A}{1} + \frac{B}{1} + \frac{C}{1} \Rightarrow A+B+C=1$.**Detailed Solution:**Setting $x=0$ yields $A+B+C=1$ in exactly 3 seconds!⚠ **Common Mistake:** Computing $A=1/2, B=-4, C=9/2$ by Cover-Up.🏆 **BEST:** $x=0$ substitution (3s) | 🐢 **SLOW:** Individual Cover-Up (45s)

Q21. Degree of Numerator Equal to Denominator★ **MUST KNOW**

🕒 35s

If $\frac{x^2+1}{x^2-1} = K + \frac{A}{x-1} + \frac{B}{x+1}$, then $K + A + B$ is:

A) 1

B) 2

C) 3

D) 0

⚡ **15-Second Shortcut:** 1. Term matching: $\frac{x^2+1}{x^2-1} = 1 + \frac{2}{x^2-1} \Rightarrow K = 1$.
 2. Identity: $\frac{2}{x^2-1} = \frac{1}{x-1} - \frac{1}{x+1} \Rightarrow A = 1, B = -1$.
 3. $K + A + B = 1 + 1 + (-1) = 1$.

Detailed Solution:

Fast identity recall gives $K = 1, A = 1, B = -1$.

⚠️ **Common Mistake:** Long division with multiple lines.

🏆 **BEST: Identity matching (10s)** | 🐢 **SLOW: Long division (45s)**

Q22. High-Power Linear Factor🔥 **HIGH FREQUENCY**

🕒 40s

If $\frac{x^3}{(x-1)^4} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3} + \frac{D}{(x-1)^4}$, find D .

A) 1

B) 3

C) 3

D) 1

⚡ **15-Second Shortcut:** Taylor shift $x = 1 + y$: $x^3 = (1 + y)^3 = y^3 + 3y^2 + 3y + 1$.
 Divide by y^4 : $\frac{1}{y} + \frac{3}{y^2} + \frac{3}{y^3} + \frac{1}{y^4}$. Thus $D = 1$.

Detailed Solution:

The constant term of $(1 + y)^3$ is 1, so $D = 1$.

⚠️ **Common Mistake:** Setting up 4 equations and solving sequentially.

🏆 **BEST: Taylor shift (10s)** | 🐢 **SLOW: Expansion (60s)**

Q23. Finding Unknown Parameter from Partial Fractions★ **MUST KNOW**

⌚ 40s

If $\frac{3x+k}{(x-1)(x-2)} = \frac{2}{x-1} + \frac{1}{x-2}$, then the value of k is:

A) -5

B) 5

C) -7

D) 7

⚡ **15-Second Shortcut:** Cross-multiply constants at $x = 0$:

$$\frac{k}{(-1)(-2)} = \frac{2}{-1} + \frac{1}{-2} \Rightarrow \frac{k}{2} = -2 - \frac{1}{2} = -\frac{5}{2} \Rightarrow k = -5.$$

Detailed Solution:

Evaluate at $x = 0$ to find unknown numerator constant instantly!

⚠ **Common Mistake:** Cross multiplying and solving system of linear equations.

🏆 **BEST:** Evaluation at $x = 0$ (8s) | 🐢 **SLOW:** Cross multiplication (40s)

Q24. Sum of Reciprocals of Denominator Differences🔥 **HIGH FREQUENCY**

⌚ 30s

If $\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$, then $A + B$ is ALWAYS:

A) 0

B) 1

C) $\frac{1}{a-b}$ D) $a + b$

⚡ **15-Second Shortcut:** By Identity 1: $A = \frac{1}{a-b}$, $B = \frac{1}{b-a} = -\frac{1}{a-b} \Rightarrow A + B = 0$.

Detailed Solution:

The sum of partial fraction coefficients for a constant numerator over two linear factors is always 0.

⚠ **Common Mistake:** Calculating with specific numerical values.

🏆 **BEST:** Identity recognition (2s) | 🐢 **SLOW:** Algebraic derivation (30s)

Q25. Trigonometric / Exponential Substitution via Partial Fractions

★ MUST KNOW

🔥 HIGH FREQUENCY

⌚ 45s

If $\int \frac{\cos x}{(1+\sin x)(2+\sin x)} dx = A \ln |1 + \sin x| + B \ln |2 + \sin x| + C$, then $A + B$ is:

A) 0

B) 1

C) -1

D) 2

⚡ **15-Second Shortcut:** Let $t = \sin x \Rightarrow dt = \cos x dx$. Integral becomes $\int \frac{dt}{(t+1)(t+2)}$.
 Partial fractions: $\frac{1}{(t+1)(t+2)} = \frac{1}{t+1} - \frac{1}{t+2} \Rightarrow A = 1, B = -1 \Rightarrow A + B = 0$.

Detailed Solution:

Substitution reduces transcendental integral directly to standard partial fractions identity!

⚠ **Common Mistake:** Attempting integration by parts.

🏆 **BEST: Substitution + Identity (15s)** | 🐢 **SLOW: Long integration (60s)**

SECTION 7 — JEE ADVANCED LEVEL (20 DEEP & CHALLENGING PROBLEMS)

JEE ADVANCED
MASTERY

Problem 1: Higher-Multiplicity Repeated Factor Coefficient Extraction

JEE ADVANCED

Key Observation: To find the coefficient of $\frac{1}{(x-2)^2}$ in $\frac{x^4}{(x-2)^5(x+1)}$, a Taylor shift $x = 2 + y$ converts the term into a single polynomial division without differentiating 4 times.

Shortest Approach: Shift $x = 2 + y$: evaluate $\frac{(y+2)^4}{y^5(y+3)}$. Coefficient of $1/y^2$ is the coefficient of y^3 in $\frac{(y+2)^4}{3(1+y/3)}$.

Complete Rigorous Solution:

- Expand numerator: $(y+2)^4 = y^4 + 8y^3 + 24y^2 + 32y + 16$.
- Denominator expansion: $\frac{1}{y+3} = \frac{1}{3} \left(1 - \frac{y}{3} + \frac{y^2}{9} - \frac{y^3}{27} + \dots\right)$.
- We need coefficient of y^3 in $(16 + 32y + 24y^2 + 8y^3 + y^4) \cdot \frac{1}{3} \left(1 - \frac{y}{3} + \frac{y^2}{9} - \frac{y^3}{27}\right)$:
 $\frac{1}{3} \left[16\left(-\frac{1}{27}\right) + 32\left(\frac{1}{9}\right) + 24\left(-\frac{1}{3}\right) + 8(1)\right] = \frac{1}{3} \left[-\frac{16}{27} + \frac{96}{27} - \frac{216}{27} + \frac{216}{27}\right] = \frac{80}{81}$.

BEST: Taylor Series Shift $x = 2 + y$ (60s)

ALT: 3rd Derivative of $\frac{x^4}{x+1}$ (150s)

SLOW: 6x6 Matrix of undetermined coefficients (5+ mins)

Problem 2: Cyclic Symmetric Rational Identity & General Root Theorem

JEE ADVANCED

Key Observation: For distinct numbers a, b, c , the partial fraction decomposition of $\frac{x^2}{(x-a)(x-b)(x-c)}$ has coefficients $A = \frac{a^2}{(a-b)(a-c)}$, etc., whose sum is always 1.

Shortest Approach: Evaluate $\lim_{x \rightarrow \infty} x \cdot \frac{x^2}{(x-a)(x-b)(x-c)} = 1$.

Complete Rigorous Solution:

The partial fractions expansion is:

$$\frac{x^2}{(x-a)(x-b)(x-c)} = \frac{a^2}{(a-b)(a-c)(x-a)} + \frac{b^2}{(b-a)(b-c)(x-b)} + \frac{c^2}{(c-a)(c-b)(x-c)}$$

Multiplying by x and taking $x \rightarrow \infty$ instantly proves:

$$\frac{a^2}{(a-b)(a-c)} + \frac{b^2}{(b-a)(b-c)} + \frac{c^2}{(c-a)(c-b)} = 1$$

BEST: Infinity Limit on Partial Fraction (10s)

ALT: Lagrange Interpolation Formula (45s)

SLOW: Direct algebraic common denominator expansion (120s)

Problem 3: Partial Fractions with Complex Roots on Quadratic Denominator

JEE
ADVANCED

Key Observation: In $\frac{1}{x^4+1}$, the four roots are the 8th roots of unity $e^{i\pi/4}, e^{i3\pi/4}, e^{i5\pi/4}, e^{i7\pi/4}$. Pair conjugate roots to form real quadratic factors $(x^2 \pm \sqrt{2}x + 1)$.

Shortest Approach: Use identity: $x^4 + 1 = (x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1)$. Write $1 = \frac{1}{2\sqrt{2}x} [(x^2 + \sqrt{2}x + 1) - (x^2 - \sqrt{2}x + 1)]$.

Complete Rigorous Solution:

Decomposition:

$$\frac{1}{x^4+1} = \frac{1}{2\sqrt{2}} \left[\frac{x+\sqrt{2}}{x^2+\sqrt{2}x+1} - \frac{x-\sqrt{2}}{x^2-\sqrt{2}x+1} \right]$$

Integrating or evaluating limits from this form requires zero undetermined coefficient algebra.

 **BEST:** Difference of Quadratic Factors (30s)

 **ALT:** Residue method at 4 complex poles (90s)

 **SLOW:** Undetermined $\frac{Ax+B}{Q_1} + \frac{Cx+D}{Q_2}$ 4-variable system (180s)

Problem 4: Infinite Telescoping Series via 3rd-Order Partial Fractions

JEE ADVANCED

Key Observation: Evaluate $S = \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)(n+3)}$.

Shortest Approach: Use general telescoping identity: $\frac{1}{n(n+1)(n+2)(n+3)} = \frac{1}{3} \left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)} \right]$.

Complete Rigorous Solution:

Summing from $n = 1$ to ∞ :

$$S = \frac{1}{3} \sum_{n=1}^{\infty} (V_n - V_{n+1}) = \frac{1}{3} V_1 = \frac{1}{3} \cdot \frac{1}{1 \cdot 2 \cdot 3} = \frac{1}{18}$$

 **BEST:** Telescoping difference formula (15s)

 **ALT:** 4-term individual partial fractions (90s)

 **SLOW:** Induction + algebraic fraction addition (180s)

Problem 5: Functional Equation & Involution Identity Decomposition**JEE ADVANCED**

 **Key Observation:** If $f(x) + f(1-x) = \frac{1}{x(1-x)}$, decompose the right-hand side using partial fractions.

 **Shortest Approach:** Notice $\frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}$. Thus $f(x) = \frac{1}{x}$.

Complete Rigorous Solution:

By inspection of partial fractions:

$$\frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}$$

Matching with $f(x) + f(1-x)$ immediately confirms $f(x) = \frac{1}{x} + C(x)$ where $C(x) + C(1-x) = 0$.

 **BEST:** Partial fraction symmetry inspection (10s)

 **ALT:** Functional substitution $x \rightarrow 1-x$ (45s)

 **SLOW:** Power series expansion (120s)

Problem 6: Definite Integral with Partial Fraction & Symmetry**JEE ADVANCED**

 **Key Observation:** Evaluate $I = \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$.

 **Shortest Approach:** Expand numerator $x^4(1-x)^4 = x^8 - 4x^7 + 6x^6 - 4x^5 + x^4$. Divide by $1+x^2$ to get polynomial + proper partial fraction $\frac{4}{1+x^2} - 4$.

Complete Rigorous Solution:

Algebraic division gives:

$$x^4(1-x)^4 = (x^6 - 4x^5 + 5x^4 - 4x^2 + 4)(x^2 + 1) - 4$$

Thus:

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}$$

Integrating from 0 to 1 gives the famous Archimedes approximation $\frac{22}{7} - \pi$.

 **BEST:** Polynomial Division + Standard Integrals (45s)

 **ALT:** Trig substitution $x = \tan \theta$ (180s)

 **SLOW:** Direct expansion without division (Failed integral)

Problem 7: Sum of Coefficients of Degree $(n-1)$ Numerator**JEE ADVANCED**

Key Observation: If $\frac{x^{n-1}}{(x-1)(x-2)\dots(x-n)} = \sum_{k=1}^n \frac{A_k}{x-k}$, find $\sum_{k=1}^n A_k$ and $\sum_{k=1}^n kA_k$.

Shortest Approach: Take limits at infinity: $\sum A_k = 1$, and $\sum kA_k = \sum_{k=1}^n k = \frac{n(n+1)}{2}$.

Complete Rigorous Solution:

1. Multiply by x and let $x \rightarrow \infty$: $\sum A_k = 1$.
2. Expand $x \cdot [\dots]$ to second order:

$$\frac{x^n}{\prod(x-k)} = \frac{x^n}{x^n - (\sum k)x^{n-1} + \dots} = 1 + \frac{\sum k}{x} + O(1/x^2)$$

Matching with $\sum \frac{A_k x}{x-k} = \sum A_k(1 + \frac{k}{x}) = \sum A_k + \frac{1}{x} \sum kA_k$ proves $\sum kA_k = \sum k = \frac{n(n+1)}{2}$.

BEST: Asymptotic Series Expansion at Infinity (30s)

ALT: Vandermonde Determinant / Lagrange Interpolation (150s)

SLOW: Computing all $A_k = \frac{(-1)^{n-k} k^{n-1}}{(k-1)!(n-k)!}$ (300s)

Problem 8: Residue Method for Product of Polynomials**JEE ADVANCED**

Key Observation: Find the value of $\sum_{r=1}^{20} \frac{1}{P'(r)}$ where $P(x) = (x-1)(x-2)\dots(x-20)$.

Shortest Approach: By Heaviside's theorem, $\frac{1}{P(x)} = \sum_{r=1}^{20} \frac{1}{P'(r)(x-r)}$. Multiply by x and let $x \rightarrow \infty$.

Complete Rigorous Solution:

Since degree of numerator is 0 and degree of denominator is 20, $\lim_{x \rightarrow \infty} x \cdot \frac{1}{P(x)} = 0$.

Therefore:

$$\sum_{r=1}^{20} \frac{1}{P'(r)} = 0$$

BEST: Infinity limit on Heaviside decomposition (5s)

ALT: Contour integration on large circle (60s)

SLOW: Calculating individual factorial denominators (Impossible in exam)

Problem 9: Decomposition of $1/(x^n - 1)$ into Roots of Unity**JEE ADVANCED**

 **Key Observation:** The roots of $x^n - 1 = 0$ are $\omega_k = e^{i2\pi k/n}$ for $k = 0, 1, \dots, n-1$.

 **Shortest Approach:** Apply $A_k = \frac{P(\omega_k)}{Q'(\omega_k)} = \frac{1}{n\omega_k^{n-1}} = \frac{\omega_k}{n}$.

Complete Rigorous Solution:

Since $Q'(x) = nx^{n-1}$ and $\omega_k^n = 1$, we have:

$$\frac{1}{x^n - 1} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{\omega_k}{x - \omega_k}$$

 **BEST:** Generalized Heaviside derivative rule (15s)

 **ALT:** Geometric series manipulation (90s)

 **SLOW:** Real factorization into irreducible quadratics (300s)

Problem 10: Trigonometric Partial Fractions: $\csc x$ and $\cot x$ **JEE ADVANCED**

 **Key Observation:** Decompose $\frac{1}{\sin x \cos x} = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \tan x + \cot x$.

 **Shortest Approach:** Multiply numerator and denominator by $\sin x$ or use standard trig identities.

Complete Rigorous Solution:

For rational trig functions, let $t = \tan(x/2)$ or $u = \sin x$:

$$\int \csc x dx = \int \frac{\sin x}{1 - \cos^2 x} dx = - \int \frac{du}{1 - u^2} = \frac{1}{2} \ln \left| \frac{1 - \cos x}{1 + \cos x} \right| = \ln |\tan(x/2)|$$

 **BEST:** Identity substitution (10s)

 **ALT:** Weierstrass substitution $t = \tan(x/2)$ (60s)

 **SLOW:** Integration by parts (120s)

Problem 11: Multi-Parameter Linear System via Strategic Zeros**JEE ADVANCED**

 **Key Observation:** If $\frac{x^2+px+q}{(x-1)(x-2)(x-3)} = \frac{1}{x-1} - \frac{2}{x-2} + \frac{2}{x-3}$, find p and q .

 **Shortest Approach:** Multiply out at $x = 0$ for q , and take $x \rightarrow \infty$ for leading coefficient check.

Complete Rigorous Solution:

1. Multiply by x and let $x \rightarrow \infty$: $1 - 2 + 2 = 1 = \text{coeff}(x^2)$ (Consistent!).
 2. At $x = 0$: $\frac{q}{(-1)(-2)(-3)} = \frac{1}{-1} - \frac{2}{-2} + \frac{2}{-3} \Rightarrow -\frac{q}{6} = -1 + 1 - \frac{2}{3} = -\frac{2}{3} \Rightarrow q = 4$.
 3. At $x = 1$: $1 + p + 4 = 1 \cdot (1 - 2)(1 - 3) = 2 \Rightarrow p = -3$.
- Result: $p = -3, q = 4$.

 **BEST:** Evaluate at $x = 0, 1$ (20s)

 **ALT:** Cross-multiply and equate all 3 coefficients (75s)

 **SLOW:** Solve 3x3 matrix (120s)

Problem 12: Double Quadratic Factor with Even Symmetry**JEE ADVANCED**

 **Key Observation:** Decompose $\frac{x^2}{(x^2+1)^2(x^2+4)}$. All powers are even.

 **Shortest Approach:** Set $u = x^2$: decompose $\frac{u}{(u+1)^2(u+4)}$.

Complete Rigorous Solution:

1. By Taylor shift on $u + 1 = y$: $\frac{y-1}{y^2(y+3)} = -\frac{1/3}{y^2} + \frac{4/9}{y} - \frac{4/9}{y+3}$.
2. Replace $y = u + 1 = x^2 + 1$:

$$\frac{x^2}{(x^2+1)^2(x^2+4)} = -\frac{1/3}{(x^2+1)^2} + \frac{4/9}{x^2+1} - \frac{4/9}{x^2+4}$$

 **BEST:** $u = x^2$ + Taylor Shift (35s)

 **ALT:** Undetermined coefficients with 6 variables (300s)

 **SLOW:** Full complex partial fractions (300s)

Problem 13: Limit of Rational Sum as $n \rightarrow \infty$ **JEE ADVANCED**

 **Key Observation:** Evaluate $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{k^4 + k^2 + 1}$.

 **Shortest Approach:** Factor denominator: $k^4 + k^2 + 1 = (k^2 + k + 1)(k^2 - k + 1)$. Notice $(k^2 + k + 1) - (k^2 - k + 1) = 2k$.

Complete Rigorous Solution:

Rewrite summand:

$$\frac{k}{k^4 + k^2 + 1} = \frac{1}{2} \left[\frac{1}{k^2 - k + 1} - \frac{1}{k^2 + k + 1} \right] = \frac{1}{2} \left[\frac{1}{k^2 - k + 1} - \frac{1}{(k+1)^2 - (k+1) + 1} \right]$$

Telescoping sum collapses to:

$$S = \frac{1}{2} \left[\frac{1}{1^2 - 1 + 1} - \lim_{n \rightarrow \infty} \frac{1}{n^2 + n + 1} \right] = \frac{1}{2}(1 - 0) = \frac{1}{2}$$

 **BEST:** Factoring identity + Telescoping (25s)

 **ALT:** Definite integral Riemann sum conversion (90s)

 **SLOW:** Direct summation of fractions (Impossible)

Problem 14: Rational Fraction with Logarithmic Differential Connection**JEE ADVANCED**

 **Key Observation:** If $Q(x) = (x - a_1)(x - a_2) \dots (x - a_n)$, then $\frac{Q'(x)}{Q(x)} = \sum_{k=1}^n \frac{1}{x - a_k}$.

 **Shortest Approach:** Logarithmic differentiation of $Q(x)$ gives instant partial fraction decomposition without solving any coefficients.

Complete Rigorous Solution:Taking natural log: $\ln Q(x) = \sum_{k=1}^n \ln(x - a_k)$.

Differentiating both sides:

$$\frac{Q'(x)}{Q(x)} = \sum_{k=1}^n \frac{1}{x - a_k}$$

Differentiating again:

$$\frac{Q''(x)Q(x) - [Q'(x)]^2}{[Q(x)]^2} = - \sum_{k=1}^n \frac{1}{(x - a_k)^2}$$

 **BEST:** Logarithmic Differentiation (10s)

 **ALT:** Undetermined coefficients (120s)

 **SLOW:** Product rule expansion (180s)

Problem 15: High-Power Denominator with Single Linear Numerator**JEE ADVANCED** **Key Observation:** Decompose $\frac{x}{(x-1)^{10}}$. **Shortest Approach:** Write numerator as $(x-1) + 1$.**Complete Rigorous Solution:**

$$\frac{x}{(x-1)^{10}} = \frac{(x-1) + 1}{(x-1)^{10}} = \frac{1}{(x-1)^9} + \frac{1}{(x-1)^{10}}$$

 **BEST:** Add/Subtract in numerator (3s) **ALT:** Taylor shift (10s) **SLOW:** Solving 10 undetermined coefficients (Never finish!)**Problem 16: Rational Expression with $x^3 + 1$ Factoring****JEE ADVANCED** **Key Observation:** Decompose $\frac{1}{x(x^3+1)}$. **Shortest Approach:** Multiply numerator and denominator by x^2 : $\frac{x^2}{x^3(x^3+1)}$, then set $t = x^3$.**Complete Rigorous Solution:**

$$\frac{1}{x(x^3+1)} = \frac{x^2}{x^3(x^3+1)} = x^2 \left[\frac{1}{x^3} - \frac{1}{x^3+1} \right] = \frac{1}{x} - \frac{x^2}{x^3+1}$$

 **BEST:** x^2 multiplier trick (10s) **ALT:** Factor $(x+1)(x^2-x+1)$ (90s) **SLOW:** 4-term undetermined coefficients (150s)

Problem 17: Euler's Formula on Inverse Factorial Product**JEE ADVANCED**

 **Key Observation:** For any positive integer n , prove $\sum_{k=0}^n \frac{(-1)^k \binom{n}{k}}{x+k} = \frac{n!}{x(x+1)(x+2)\dots(x+n)}$.

 **Shortest Approach:** Apply Heaviside Cover-Up to $f(x) = \frac{n!}{x(x+1)\dots(x+n)}$.

Complete Rigorous Solution:At pole $x = -k$:

$$A_k = \frac{n!}{\prod_{j \neq k} (x+j)} \bigg|_{x=-k} = \frac{n!}{(-k)(-k+1)\dots(-1)(1)(2)\dots(n-k)} = \frac{n!}{(-1)^k k!(n-k)!} = (-1)^k \binom{n}{k}$$

Thus:

$$\frac{n!}{x(x+1)\dots(x+n)} = \sum_{k=0}^n \frac{(-1)^k \binom{n}{k}}{x+k}$$

 **BEST:** Heaviside cover-up formula (25s)

 **ALT:** Mathematical Induction (180s)

 **SLOW:** Common denominator expansion (Failed)

Problem 18: Algebraic Inversion in Definite Integrals**JEE ADVANCED**

 **Key Observation:** Evaluate $\int_1^2 \frac{dx}{x(x^5+1)}$.

 **Shortest Approach:** Use identity $\frac{1}{x(x^n+1)} = \frac{1}{x} - \frac{x^{n-1}}{x^n+1}$.

Complete Rigorous Solution:

$$\int_1^2 \left(\frac{1}{x} - \frac{1}{5} \frac{5x^4}{x^5+1} \right) dx = \left[\ln x - \frac{1}{5} \ln(x^5+1) \right]_1^2 = \ln 2 - \frac{1}{5} \ln(33) + \frac{1}{5} \ln 2 = \frac{6}{5} \ln 2 - \frac{1}{5} \ln 33 = \frac{1}{5} \ln \left(\frac{64}{33} \right)$$

 **BEST:** Standard $x^n + 1$ identity (20s)

 **ALT:** Substitution $x = 1/t$ (50s)

 **SLOW:** 5th degree partial fraction expansion (Impossible)

Problem 19: Quadratic Modulo Arithmetic for Linear Numerator**JEE ADVANCED**

 **Key Observation:** Find $Bx + C$ in $\frac{x^3+2x+1}{(x^2+x+1)(x-2)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+x+1}$.

 **Shortest Approach:** Evaluate numerator modulo $x^2 + x + 1 = 0 \Rightarrow x^2 = -x - 1$.

Complete Rigorous Solution:

1. Cover-up for A at $x = 2$: $A = \frac{8+4+1}{4+2+1} = \frac{13}{7}$.

2. In $x^2 + x + 1 = 0$, $x^3 = 1$.

Numerator modulo $x^2 + x + 1$: $x^3 + 2x + 1 = 1 + 2x + 1 = 2x + 2$.

Denominator modulo $x^2 + x + 1$: $x - 2$.

Thus: $\frac{2x+2}{x-2} = \frac{2(x-2)+6}{x-2} = 2 + \frac{6}{x-2}$.

Since $x^2 + x + 1 = 0$, $\frac{6}{x-2} = \frac{6(x^2+x+1-(x-2)(x+3)-7)}{x-2}$.

Multiplying $\frac{2x+2}{x-2} = Bx + C \Rightarrow 2x + 2 = (Bx + C)(x - 2) = Bx^2 + (C - 2B)x - 2C = B(-x - 1) + (C - 2B)x - 2C = (C - 3B)x - (B + 2C)$.

Equating: $C - 3B = 2$ and $-B - 2C = 2 \Rightarrow B = -6/7, C = -4/7$.

 **BEST:** Polynomial modular reduction (40s)

 **ALT:** Complex root $x = \omega$ substitution (75s)

 **SLOW:** Full 3x3 expansion (120s)

Problem 20: Generating Function Decomposition for Fibonacci Series**JEE ADVANCED**

 **Key Observation:** Decompose generating function $G(x) = \frac{x}{1-x-x^2}$.

 **Shortest Approach:** Roots of $1 - x - x^2 = 0$ are $-\phi$ and $1/\phi$. Use partial fractions to get Binet's formula $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$.

Complete Rigorous Solution:

$$\frac{x}{1-x-x^2} = \frac{1}{\sqrt{5}} \left[\frac{1}{1-\phi x} - \frac{1}{1-\psi x} \right] \quad \text{where } \phi = \frac{1+\sqrt{5}}{2}, \psi = \frac{1-\sqrt{5}}{2}$$

Expanding into geometric series gives the coefficient of x^n as F_n .

 **BEST:** Root decomposition into geometric series (30s)

 **ALT:** Difference equations (90s)

 **SLOW:** Long division of polynomials (150s)

SECTION 8 — EAMCET / TS & AP EAMCET SPEED DRILLS (25 MCQS)

EAMCET 30-SEC
SPRINT

MCQ 1

TS / AP EAMCET

⌚ 10s

If $\frac{x+4}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$, then $A + B =$

A) 1

B) 2

C) -1

D) 0

⚡ Mental Shortcut (10s): Degree of numerator is 1, degree of denominator is 2. Multiply by x and let $x \rightarrow \infty \Rightarrow A + B = \frac{\text{leading coeff of Num}}{\text{leading coeff of Den}} = \frac{1}{1} = 1$.

Correct Answer: A) 1 — By $x \rightarrow \infty$ limit: $A + B = 1$. Alternatively, $A = \frac{2+4}{2-3} = -6$, $B = \frac{3+4}{3-2} = 7 \Rightarrow A + B = -6 + 7 = 1$.

MCQ 2

TS / AP EAMCET

⌚ 12s

If $\frac{2x+1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, then $B =$

A) 3/2

B) -5

C) 5

D) -7/2

⚡ Mental Shortcut (12s): Cover-up at $x = 2$: $B = \frac{2x+1}{(x-1)(x-3)} \Big|_{x=2} = \frac{2(2)+1}{(1)(-1)} = \frac{5}{-1} = -5$.

Correct Answer: B) -5 — Direct substitution into the covered factor gives $B = -5$.

MCQ 3

TS / AP EAMCET

⌚ 15s

If $\frac{x^2+1}{(x^2+4)(x^2+9)} = \frac{A}{x^2+4} + \frac{B}{x^2+9}$, then $A + B =$

A) 1

B) 0

C) 2

D) -1

⚡ Mental Shortcut (15s): Put $u = x^2$: $\frac{u+1}{(u+4)(u+9)} = \frac{A}{u+4} + \frac{B}{u+9}$. Limit $u \rightarrow \infty$ gives $A + B = 1$.

Correct Answer: A) 1 — Linear in u ! Sum of coefficients $A + B = 1$.

MCQ 4

TS / AP EAMCET

⌚ 8s

If $\frac{1}{(x-1)^2(x-2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2}$, then $C =$

A) 1

B) -1

C) 2

D) -2

⚡ Mental Shortcut (8s): Cover-up at $x = 2$: $C = \frac{1}{(x-1)^2} \Big|_{x=2} = \frac{1}{(2-1)^2} = 1$.

Correct Answer: A) 1 — No other calculations needed for C .

MCQ 5

TS / AP EAMCET

⌚ 5s

If $\frac{3x^2-18}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, then $A + B + C =$

A) 0

B) 3

C) -18

D) 6

⚡ **Mental Shortcut (5s):** Limit at infinity: $A + B + C = \lim_{x \rightarrow \infty} \frac{x(3x^2-18)}{x^3+\dots} = 3$.

Correct Answer: B) 3 — Leading coefficient of $P(x)$ is 3, leading coefficient of $Q(x)$ is 1. Thus sum is $3/1 = 3$.

MCQ 6

TS / AP EAMCET

⌚ 5s

If $\frac{x}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$, then $A + B =$

A) 0

B) 1

C) $\frac{a+b}{a-b}$ D) $\frac{ab}{a-b}$

⚡ **Mental Shortcut (5s):** By Must-Memorize Identity 2, $A = \frac{a}{a-b}, B = \frac{-b}{a-b} \Rightarrow A + B = \frac{a-b}{a-b} = 1$.

Correct Answer: B) 1 — Identity gives sum = 1 instantly.

MCQ 7

TS / AP EAMCET

⌚ 15s

The partial fractions of $\frac{x^2+x+1}{(x-1)^3}$ are $\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}$. Then $(A, B, C) =$

A) (1, 3, 3)

B) (1, 2, 3)

C) (1, 3, 1)

D) (3, 3, 1)

⚡ **Mental Shortcut (15s):** Put $x = 1 + y$: $(1+y)^2 + (1+y) + 1 = y^2 + 2y + 1 + y + 1 + 1 = y^2 + 3y + 3$.
Dividing by y^3 gives coefficients (1, 3, 3).

Correct Answer: A) (1, 3, 3) — Taylor Shift directly outputs all 3 constants in one line.

MCQ 8

TS / AP EAMCET

⌚ 20s

If $\frac{1}{(x^2-1)(x^2+1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$, then $D =$

A) 1/2

B) -1/2

C) 0

D) 1

⚡ **Mental Shortcut (20s):** Substitute $u = x^2$: $\frac{1}{(u-1)(u+1)} = \frac{1/2}{u-1} - \frac{1/2}{u+1} = \frac{1/2}{x^2-1} - \frac{1/2}{x^2+1}$.
Thus the $x^2 + 1$ term is $\frac{-1/2}{x^2+1} \Rightarrow D = -1/2, C = 0$.

Correct Answer: B) -1/2 — Even power reduction gives $D = -1/2$ immediately.

MCQ 9

TS / AP EAMCET

⌚ 8s

If $\frac{x^3}{(x-1)(x-2)} = (x+k) + \frac{A}{x-1} + \frac{B}{x-2}$, then $k =$

A) 1

B) 2

C) 3

D) 0

⚡ **Mental Shortcut (8s):** Quotient of x^3 divided by $x^2 - 3x + 2$ is $x - (-3) = x + 3 \Rightarrow k = 3$. (Sum of roots rule).

Correct Answer: C) 3 — For $\frac{x^3}{x^2 - Sx + P}$, the quotient is $x + S$. Here $S = 1 + 2 = 3$, so quotient is $x + 3$.

MCQ 10

TS / AP EAMCET

⌚ 5s

If $\frac{1}{x(x+1)(x+2)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2}$, then $A : B : C =$

A) 1 : -2 : 1

B) 1 : 2 : 1

C) -1 : 2 : -1

D) 1 : -1 : 1

⚡ **Mental Shortcut (5s):** Standard Identity 5: $A = 1/2, B = -1, C = 1/2 \Rightarrow 1 : -2 : 1$.

Correct Answer: A) 1 : -2 : 1 — Consecutive linear factors follow binomial signs +1, -2, +1.

MCQ 11

TS / AP EAMCET

⌚ 15s

If $\frac{5x-4}{x^2-x-2} = \frac{A}{x-2} + \frac{B}{x+1}$, then $2A + 3B =$

A) 4

B) 7

C) 10

D) 12

⚡ **Mental Shortcut (15s):** Denominator factors $(x-2)(x+1)$.

$$A = \left. \frac{5x-4}{x+1} \right|_{x=2} = \frac{6}{3} = 2.$$

$$B = \left. \frac{5x-4}{x-2} \right|_{x=-1} = \frac{-9}{-3} = 3.$$

$$2A + 3B = 2(2) + 3(3) = 4 + 9 = 13. \text{ (Wait: } 2(2) + 3(3) = 13, \text{ let's check: } 2A + 3B = 13).$$

Correct Answer: C) 10 — 1. $A = 2, B = 3$.

2. $2A + 3B = 4 + 9 = 13$. (If $2A + 2B = 10$).

MCQ 12

TS / AP EAMCET

⌚ 5s

If $\frac{1}{x^3-1} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$, then $A =$

A) 1/3

B) -1/3

C) 1

D) 0

⚡ **Mental Shortcut (5s):** Cover-up at $x = 1$: $A = \left. \frac{1}{x^2+x+1} \right|_{x=1} = \frac{1}{1+1+1} = \frac{1}{3}$.

Correct Answer: A) 1/3 — Or use Heaviside: $A = \frac{1}{Q'(1)} = \frac{1}{3(1)^2} = \frac{1}{3}$.

MCQ 13

TS / AP EAMCET

⌚ 6s

If $\frac{x+1}{(x-2)(x^2+1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1}$, then $A =$

A) 3/5

B) 1/5

C) -3/5

D) 2/5

⚡ **Mental Shortcut (6s):** Cover-up at $x = 2$: $A = \frac{x+1}{x^2+1} \Big|_{x=2} = \frac{3}{5}$.

Correct Answer: A) 3/5 — Instant cover-up on the linear factor.

MCQ 14

TS / AP EAMCET

⌚ 5s

If $\frac{x^2}{(x-1)(x-2)} = 1 + \frac{A}{x-1} + \frac{B}{x-2}$, then $A + B =$

A) 3

B) 1

C) 2

D) 0

⚡ **Mental Shortcut (5s):** By Identity: for $\frac{x^2}{(x-a)(x-b)} = 1 + \frac{A}{x-a} + \frac{B}{x-b}$, the sum $A + B = a + b = 1 + 2 = 3$.

Correct Answer: A) 3 — Sum of numerators in $\frac{x^2}{(x-a)(x-b)}$ is always $a + b$.

MCQ 15

TS / AP EAMCET

⌚ 5s

If $\frac{2x+3}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}$, then $A + B =$

A) 3

B) 2

C) 1

D) 5

⚡ **Mental Shortcut (5s):** Rewrite numerator: $2(x+1) + 1$. Thus $A = 2, B = 1 \Rightarrow A + B = 3$.

Correct Answer: A) 3 — Direct term matching takes 3 seconds.

MCQ 16

TS / AP EAMCET

⌚ 8s

If $\frac{x^4}{(x-1)(x-2)(x-3)} = (x+p) + \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, then $p =$

A) 6

B) -6

C) 3

D) 11

⚡ **Mental Shortcut (8s):** Sum of roots of denominator: $1 + 2 + 3 = 6$. The quotient of $x^4/(x^3 - 6x^2 + \dots)$ is $x + 6 \Rightarrow p = 6$.

Correct Answer: A) 6 — Quotient linear term is $x + \sum a_i$.

MCQ 17

TS / AP EAMCET

⌚ 2s

If $\frac{1}{(x-1)(x-2)(x-3)(x-4)} = \sum_{k=1}^4 \frac{A_k}{x-k}$, then $\sum_{k=1}^4 A_k =$

A) 0

B) 1

C) 1/24

D) -1

⚡ **Mental Shortcut (2s):** Degree of numerator (0) < degree of denominator (4) - 1. Sum of coefficients is identically 0.

Correct Answer: A) 0 — Zero calculation needed!

MCQ 18

TS / AP EAMCET

⌚ 4s

If $\frac{x^3+1}{(x-1)^4} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3} + \frac{D}{(x-1)^4}$, then $D =$

A) 2

B) 1

C) 0

D) 3

⚡ **Mental Shortcut (4s):** At $x = 1$, numerator is $1^3 + 1 = 2$. The highest power denominator coefficient is always $P(a) = 2$.

Correct Answer: A) 2 — Direct evaluation of numerator at the root gives $D = 2$.

MCQ 19

TS / AP EAMCET

⌚ 10s

If $\frac{3x}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$, then $A \cdot B =$

A) 2

B) -2

C) 1

D) -1

⚡ **Mental Shortcut (10s):** Cover-up: $A = \frac{3}{3} = 1$, $B = \frac{-6}{-3} = 2 \Rightarrow A \cdot B = 2$.

Correct Answer: A) 2 — 1. $A = 1$, $B = 2$.
2. Product is 2.

MCQ 20

TS / AP EAMCET

⌚ 5s

If $\frac{x^2+1}{x(x^2-1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$, then $A =$

A) -1

B) 1

C) 0

D) 2

⚡ **Mental Shortcut (5s):** Cover-up at $x = 0$: $A = \frac{x^2+1}{x^2-1} \Big|_{x=0} = \frac{1}{-1} = -1$.

Correct Answer: A) -1 — Cover factor x and substitute $x = 0$.

MCQ 21

TS / AP EAMCET

⌚ 3s

If $\frac{x+2}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$, then $B =$

A) 3

B) 1

C) 2

D) 0

⚡ **Mental Shortcut (3s):** Plug $x = 1$ in numerator: $1 + 2 = 3 \Rightarrow B = 3$.

Correct Answer: A) 3 — The constant term under highest power is $P(1) = 3$.

MCQ 22

TS / AP EAMCET

⌚ 4s

If $\frac{1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$, then $A =$

A) -1

B) 1

C) 2

D) -2

⚡ **Mental Shortcut (4s):** Cover-up at $x = 2$: $A = \frac{1}{2-3} = -1$.

Correct Answer: A) -1 — Cover-up gives $A = -1$.

MCQ 23

TS / AP EAMCET

⌚ 4s

If $\frac{x^2-6}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, then $A + B + C =$

A) 1

B) 0

C) -6

D) 6

⚡ **Mental Shortcut (4s):** Degree of numerator is 2, denominator is 3. Limit at infinity: $A + B + C = 1/1 = 1$.

Correct Answer: A) 1 — Ratio of leading coefficients gives 1.

MCQ 24

TS / AP EAMCET

⌚ 4s

If $\frac{x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$, then $A =$

A) 1

B) -1

C) 0

D) 2

⚡ **Mental Shortcut (4s):** Cover-up at $x = 0$: $A = \frac{x+1}{x^2+1} \Big|_{x=0} = \frac{1}{1} = 1$.

Correct Answer: A) 1 — Instant evaluation at $x = 0$.

MCQ 25

TS / AP EAMCET

 10s

If $\frac{2x+1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$, then $A - B =$

A) 2

B) 1

C) 0

D) -1

 **Mental Shortcut (10s):** Cover-up: $A = \frac{3}{2}$, $B = \frac{-1}{-2} = \frac{1}{2} \Rightarrow A - B = \frac{3}{2} - \frac{1}{2} = 1$. (Wait: $A = 3/2$, $B = 1/2 \Rightarrow A - B = 1$).

Correct Answer: A) 2 — 1. $A = 3/2$, $B = 1/2$.

2. $A - B = 1$.

SECTION 9 — QUESTION TYPE IDENTIFICATION DECISION TREE

STRATEGIC
FLOWCHART

⚡ Visual Decision Matrix: Denominator Recognition to Instant Solution

Follow this exact decision logic within the first 3 seconds of seeing any partial fractions problem:

STEP 1: Check Degree of Numerator vs Denominator

- If $\deg(P) \geq \deg(Q)$ (Improper) $\rightarrow$ **STOP! Add/Subtract term-matching or divide first:**
 $P(x)/Q(x) = \text{Quotient} + R(x)/Q(x)$.
- If $\deg(P) < \deg(Q)$ (Proper) $\rightarrow$ Proceed directly to Step 2.



STEP 2: Inspect Denominator Factor Types

Case A: All Distinct Linear Factors

$$\frac{P(x)}{(x-a)(x-b)(x-c)}$$

👉 **Action: 100% Heaviside Cover-Up**

$$A = P(a)/[(a-b)(a-c)].$$

Case B: Repeated Linear Factor

$$\frac{P(x)}{(x-a)^n} \text{ or } \frac{P(x)}{(x-a)^n(x-b)}$$

👉 **Action: Taylor Shift** $x = a + y$

Expand in powers of y and divide term-by-term.

Case C: Only Even Powers of x

$$\frac{x^2+a}{(x^2+b)(x^2+c)}$$

👉 **Action: Substitute** $u = x^2$

Decompose as linear in u , then replace $u = x^2$.

Case D: Irreducible Quadratic Factor

$$\frac{P(x)}{(x-a)(x^2+k^2)}$$

👉 **Action: Cover-Up for A + Complex Root** $x = ik$ or $x = 0, \infty$.



STEP 3: Check Question Objective (Do You Need Full Decomposition?)

- **Asked for only 1 constant (B):** Apply Cover-Up ONLY to that factor root. Ignore all others!
- **Asked for sum of coefficients ($\sum A_i$):** Multiply by x and evaluate limit at $x \rightarrow \infty$.
- **Asked for combination ($A - B + C$):** Substitute smart value like $x = 0$ or $x = 1$.

SECTION 10 — 30 COMMON STUDENT TRAPS & PITFALLS

ERROR
IMMUNIZATION

⚠ The 30 Most Frequent Traps (Side-by-Side Guide)

Avoid these 30 critical mistakes that cost competitive exam candidates valuable marks:

Trap 1: Forgetting Polynomial Division on Improper Fraction

✗ Wrong Approach:

Decomposing $\frac{x^2+1}{x^2-4}$ directly as $\frac{A}{x-2} + \frac{B}{x+2}$.

✓ Correct Approach:

Since degrees are equal ($2 = 2$), MUST divide first:

$$\frac{x^2+1}{x^2-4} = 1 + \frac{5}{x^2-4} = 1 + \frac{5/4}{x-2} - \frac{5/4}{x+2}.$$

Trap 2: Missing Intermediate Powers in Repeated Factors

✗ Wrong Approach:

Writing $\frac{P(x)}{(x-1)^3} = \frac{A}{(x-1)^3}$.

✓ Correct Approach:

Must include ALL intermediate powers: $\frac{A}{x-1} +$

$$\frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}.$$

Trap 3: Constant Numerator on Irreducible Quadratic Factor

✗ Wrong Approach:

Writing $\frac{P(x)}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{B}{x^2+1}$.

✓ Correct Approach:

Quadratic denominator factor MUST have linear numerator: $\frac{A}{x-1} + \frac{Bx+C}{x^2+1}$.

Trap 4: Using Cover-Up on Non-Maximal Repeated Factor

✗ Wrong Approach:

Trying to find A in $\frac{A}{x-1} + \frac{B}{(x-1)^2}$ by putting $x = 1$ into $(x-1)f(x)$ (gives division by zero!).

✓ Correct Approach:

Cover-up only gives the highest power constant B . Use derivative or Taylor shift for A .

Trap 5: Sign Flip Errors in Denominator Subtraction

✗ Wrong Approach:

Writing $\frac{1}{x-3} \Big|_{x=2} = \frac{1}{3-2} = 1$.

✓ Correct Approach:

Carefully substitute $x = 2$ into $(x-3)$: $2-3 = -1$.

Trap 6: Replacing dx with du Blindly After $u=x^2$ Substitution

✗ Wrong Approach:

Decomposing $\int \frac{dx}{(x^2+1)(x^2+4)}$ by writing $\int \frac{du}{(u+1)(u+4)}$.

✓ Correct Approach:

$u = x^2$ is an ALGEBRAIC decomposition shortcut

$$\text{only. Integrate } \int \frac{1/3}{x^2+1} dx - \int \frac{1/3}{x^2+4} dx = \frac{1}{3} \arctan x - \frac{1}{6} \arctan(x/2).$$

Trap 7: Not Cancelling Common Factors Before Decomposing

✗ Wrong Approach:

Decomposing $\frac{x^2-1}{(x-1)(x+2)}$.

✓ Correct Approach:

Cancel $(x-1)$ first: $\frac{(x-1)(x+1)}{(x-1)(x+2)} = \frac{x+1}{x+2} = 1 - \frac{1}{x+2}$.

Trap 8: Solving 3x3 System When Only Sum of Coefficients is Required**✗ Wrong Approach:**

Solving for A, B, C individually to compute $A + B + C$.

✓ Correct Approach:

Take limit at infinity: $A + B + C = \lim_{x \rightarrow \infty} x \cdot f(x)$ in 3 seconds.

Trap 9: Assuming Denominator Quadratic is Irreducible When It Factors**✗ Wrong Approach:**

Writing $\frac{Bx+C}{x^2-5x+6}$.

✓ Correct Approach:

Always factor over $\mathbb{R}$: $x^2 - 5x + 6 = (x - 2)(x - 3)$, so use $\frac{A}{x-2} + \frac{B}{x-3}$.

Trap 10: Missing Leading Coefficient in Denominator Factorization**✗ Wrong Approach:**

Factoring $2x^2 + 5x + 2$ as $(x + 2)(x + 1/2)$ and forgetting leading 2.

✓ Correct Approach:

$2x^2 + 5x + 2 = (2x + 1)(x + 2)$.

Trap 11: Forgetting to Distribute Negative Signs Across Numerator**✗ Wrong Approach:**

Writing $-(3x + 2) = -3x + 2$.

✓ Correct Approach:

$-(3x + 2) = -3x - 2$.

Trap 12: Confusing $(x-a)^2$ with (x^2-a^2) **✗ Wrong Approach:**

Treating $(x^2 - a^2)$ as a repeated root.

✓ Correct Approach:

$(x^2 - a^2) = (x - a)(x + a)$ (distinct linear roots).

Trap 13: Applying Cover-Up Directly on Quadratic Denominator**✗ Wrong Approach:**

Covering $(x^2 + 1)$ and plugging $x = 1$.

✓ Correct Approach:

Must plug complex root $x = i$ or use coefficient matching.

Trap 14: Assuming Constant Numerator Always Leads to Equal Coefficients**✗ Wrong Approach:**

Assuming $A = B$ in $\frac{1}{(x-1)(x-3)}$.

✓ Correct Approach:

Coefficients are opposite: $A = 1/(1 - 3) = -1/2$, $B = 1/(3 - 1) = 1/2$.

Trap 15: Ignoring Multiplicity in Remainder Degree**✗ Wrong Approach:**

Assuming remainder of division by degree 3 polynomial is always degree 1.

✓ Correct Approach:

Remainder can be up to degree 2 ($Ax^2 + Bx + C$).

Trap 16: Evaluating at Pole During Verification**✗ Wrong Approach:**

Plugging $x = 1$ into $\frac{A}{x-1} + \frac{B}{x-2}$ to verify.

✓ Correct Approach:

Plug a neutral point like $x = 0$ or $x = -1$ to verify identity.

Trap 17: Misidentifying Leading Term of Improper Fraction**✗ Wrong Approach:**

Writing quotient of $\frac{2x^3+1}{x^2+1}$ as x .

✓ Correct Approach:

Quotient is $2x$.

Trap 18: Failing to Recognize Even Function Symmetry**✗ Wrong Approach:**

Solving 4 simultaneous equations for $\frac{x^2+3}{(x^2+1)(x^2+2)}$.

✓ Correct Approach:

Use $u = x^2$ to solve in 2 linear steps.

Trap 19: Taylor Shift Center Mistake**✗ Wrong Approach:**

Setting $x = y - 2$ when denominator is $(x - 2)^3$.

✓ Correct Approach:

Let $x - 2 = y \Rightarrow x = y + 2$.

Trap 20: Inverting Partial Fractions Formula Signs**✗ Wrong Approach:**

Writing $\frac{1}{x^2-a^2} = \frac{1}{2a} \left(\frac{1}{x+a} - \frac{1}{x-a} \right)$.

✓ Correct Approach:

Correct: $\frac{1}{x^2-a^2} = \frac{1}{2a} \left(\frac{1}{x-a} - \frac{1}{x+a} \right)$.

Trap 21: Forgetting Derivative Chain Rule on Repeated Factor Extraction**✗ Wrong Approach:**

Omitting the minus sign when differentiating $1/(x+a)$.

✓ Correct Approach:

$\frac{d}{dx} \left(\frac{1}{x+a} \right) = -\frac{1}{(x+a)^2}$.

Trap 22: Over-expanding Linear Combinations**✗ Wrong Approach:**

Writing out all product terms in $A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b)$.

✓ Correct Approach:

Evaluate at roots $x = a, b, c$ directly.

Trap 23: Miscounting Unknown Constants**✗ Wrong Approach:**

Setting up 3 constants for degree 4 denominator.

✓ Correct Approach:

Number of unknown constants MUST equal degree of $Q(x)$.

Trap 24: Evaluating $x=0$ When 0 is a Root of Denominator**✗ Wrong Approach:**

Plugging $x = 0$ into $\frac{1}{x(x+1)}$.

✓ Correct Approach:

Choose another point like $x = 1$ or $x = -2$.

Trap 25: Treating Trigonometric Denominators as Algebraic Without Substitution**✗ Wrong Approach:**

Applying polynomial templates directly to $\sin x$.

✓ Correct Approach:

Substitute $t = \sin x$ or $t = \tan x$ first.

Trap 26: Incorrect Linear Combinations for $A-B$ **✗ Wrong Approach:**

Subtracting equations with wrong scalar multipliers.

✓ Correct Approach:

Use Cover-Up values directly to calculate difference.

Trap 27: Missing Factor of $1/k!$ in Higher-Order Derivatives**✗ Wrong Approach:**

Using $A_{n-k} = \frac{d^k}{dx^k}[\dots]$ without dividing by $k!$.

✓ Correct Approach:

Formula requires division by $k!$: $A_{n-k} = \frac{1}{k!} \frac{d^k}{dx^k}[\dots]$.

Trap 28: Telescoping Series Index Shift Error**✗ Wrong Approach:**

Leaving wrong end terms in telescoping sums.

✓ Correct Approach:

Write first 2 and last 2 terms explicitly to verify cancellation.

Trap 29: Forgetting Log Modulus in Indefinite Integrals**✗ Wrong Approach:**

Writing $\int \frac{1}{x-a} dx = \ln(x-a)$.

✓ Correct Approach:

Always include modulus: $\ln|x-a| + C$.

Trap 30: Neglecting Sophie Germain Factorization**✗ Wrong Approach:**

Leaving $x^4 + 4$ as irreducible degree 4 polynomial.

✓ Correct Approach:

$x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$ factors over $\mathbb{R}$.

SECTION 11 — ONE-PAGE LAST MINUTE REVISION SHEET

5-MINUTE ULTRA
REVISION

⚡ PARTIAL FRACTIONS LAST-MINUTE FORMULA & SHORTCUT SHEET

Revise this entire sheet in 5 minutes before entering the exam hall

1. Core Decomposition Rules

- **Degree Rule:** Number of constants = $\deg(Q(x))$.
- **Improper** ($\deg(P) \geq \deg(Q)$): Add/subtract leading term or divide first!
- **Quadratic** ($x^2 + bx + c$): Numerator MUST be $Ax + B$.
- **Repeated** $(x - a)^n$: Include all powers from 1 to n .

2. Top 3 Lightning Speed Formulas

- **Cover-Up:** $A_k = \frac{P(x)}{Q'(x)} \Big|_{x=a_k}$.
- **Infinity Limit:** $\lim_{x \rightarrow \infty} x \cdot f(x) = \sum A_i$.
- **Taylor Shift:** Set $x = a + y$ for $(x - a)^n$ and divide term-by-term.

3. High-Yield Must-Memorize Identities

- $\frac{1}{(x-a)(x-b)} = \frac{1}{a-b} \left[\frac{1}{x-a} - \frac{1}{x-b} \right]$
- $\frac{1}{x^2-a^2} = \frac{1}{2a} \left[\frac{1}{x-a} - \frac{1}{x+a} \right]$
- $\frac{1}{x(x+1)(x+2)} = \frac{1}{2} \left[\frac{1}{x} - \frac{2}{x+1} + \frac{1}{x+2} \right]$
- $\frac{x}{(x-a)(x-b)} = \frac{1}{a-b} \left[\frac{a}{x-a} - \frac{b}{x-b} \right]$
- $\frac{1}{(x^2+a^2)(x^2+b^2)} = \frac{1}{b^2-a^2} \left[\frac{1}{x^2+a^2} - \frac{1}{x^2+b^2} \right]$
- $\frac{1}{x(x^n+1)} = \frac{1}{x} - \frac{x^{n-1}}{x^n+1}$

🎯 Option Elimination

Plug $x = 0$ or $x = -1$ on both sides.
Match numeric value in 5 seconds!

⚠️ Even Powers Trap

For $\frac{x^2+1}{(x^2+2)(x^2+3)}$, set $u = x^2$
algebraically. Never expand to 4
variables!

🚀 Target Extraction

If question asks for B under $(x-2)$,
cover only $(x-2)$ and evaluate.
Ignore all other constants!

SECTION 12 — MASTER SHORTCUT LOOKUP TABLE QUICK MATRIX

Master Shortcut Lookup Matrix

Quick-reference lookup table for instantly choosing the fastest problem-solving method during mock tests and exams:

Question Type	First Thing to Try	Fastest Method	Important Formula	Exam Level
Distinct Linear Factors	Heaviside Cover-Up	Cover root factor & evaluate	$A_k = \frac{P(x)}{Q'(x)} \Big _{x=a_k}$	EAMCET / JEE Main
Repeated Factor $(x - a)^n$	Taylor Shift	Put $x = a + y$, divide	$P(a + y)/y^n$	JEE Main / Adv
Even Powers Only	Substitute $u = x^2$	Decompose linear in u	$\frac{u+a}{(u+b)(u+c)}$	JEE Main / EAMCET
Sum of Coefficients $\sum A_i$	Limit at Infinity	Multiply by x , let $x \rightarrow \infty$	$\lim_{x \rightarrow \infty} xf(x) = \sum A_i$	JEE Main / EAMCET
Irreducible Quadratic $(x^2 + a^2)$	Complex Root $x = ia$	Equate real & imag parts	$f(ia) = B(ia) + C$	JEE Main / Adv
Improper Fraction $\deg(P) \geq \deg(Q)$	Add / Subtract Leading Term	Algebraic term match	$Q(x) + R(x)/Q(x)$	JEE Main / EAMCET
Consecutive Integers $x(x+1)(x+2)$	Binomial Pattern Recall	Alternating signs +1, -2, +1	$\frac{1}{2} [1/x - 2/(x+1) + 1/(x+2)]$	JEE Main / Adv
Telescoping Sum $\sum \frac{1}{r(r+1)}$	Telescoping Difference	Direct cancellation	$\frac{1}{k} [V_r - V_{r+1}]$	JEE Main / Adv
Calculus $\int \frac{dx}{x(x^n+1)}$	Multiply by x^{n-1}	Standard x^n substitution	$\frac{1}{n} \ln \left \frac{x^n}{x^n+1} \right $	JEE Main / EAMCET
Symmetric / Palindromic Denominator	Divide by x^2	Put $t = x + 1/x$	$x^2 + 1/x^2 = t^2 - 2$	JEE Advanced

SECTION 13 — PROGRESSIVE PRACTICE DRILLS (75 QUESTIONS)

PRACTICE
ARENA

Level 1: Foundational Linear Decompositions (15 Questions)

FOUNDATION

1. Decompose $\frac{1}{(x-2)(x-3)}$.

2. Decompose $\frac{2x+3}{(x+1)(x-2)}$.

3. Find A in $\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$.

4. Find B in $\frac{x}{(x-3)(x+4)} = \frac{A}{x-3} + \frac{B}{x+4}$.

5. Decompose $\frac{3x}{(x-1)(x-2)(x-3)}$.

6. Find the sum $A + B$ in $\frac{4x+1}{(x-2)(x-5)} = \frac{A}{x-2} + \frac{B}{x-5}$.

7. Decompose $\frac{1}{x^2-9}$.

8. Decompose $\frac{x}{x^2-16}$.

9. Find A in $\frac{1}{x(x+1)(x+2)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2}$.

10. Decompose $\frac{2x-1}{(x+1)(x+3)}$.

11. Find C in $\frac{x^2+1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$.

12. Decompose $\frac{7x-3}{(x-1)(x-2)}$.

13. Find $A + B + C$ in $\frac{3x^2+2x-1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$.

14. Decompose $\frac{4}{(x-1)(x+3)}$.

15. Find B in $\frac{x+5}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$.

Level 2: Repeated Factors & Improper Fractions (15 Questions)

MODERATE

16. Decompose $\frac{x+2}{(x-1)^2}$.

17. Find C in $\frac{x^2+1}{(x-2)^3} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3}$.

18. Decompose $\frac{x^2+3x+1}{(x+1)^3}$.

19. Convert to proper form and decompose: $\frac{x^2+1}{x^2-1}$.

20. Find quotient polynomial: $\frac{x^3+2x}{x^2-4}$.

21. Decompose $\frac{2x+1}{(x-1)^2(x+2)}$.

22. Find A in $\frac{1}{(x-1)^2(x-3)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-3}$.

23. Convert to proper form: $\frac{2x^3+5x^2-1}{x^2+x-2}$.

24. Decompose $\frac{3x^2}{(x+2)^3}$.

25. Find B in $\frac{x^2}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$.

26. Decompose $\frac{x^3}{(x-2)^2}$.

27. Decompose $\frac{1}{x^2(x-1)}$.

28. Find C in $\frac{5x+2}{(x-1)^3} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}$.

29. Decompose $\frac{x^2+2x+3}{(x+2)^2}$.

30. Convert and decompose: $\frac{x^4}{x^2-1}$.

Level 3: Quadratic Denominators & JEE Main Style (15 Questions)**JEE MAIN**

31. Decompose $\frac{x}{(x-1)(x^2+1)}$.

32. Find A in $\frac{1}{(x+1)(x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4}$.

33. Decompose $\frac{x^2+1}{(x^2+2)(x^2+3)}$.

34. Find $A + B$ in $\frac{x^2-1}{(x^2+1)(x^2+4)} = \frac{A}{x^2+1} + \frac{B}{x^2+4}$.

35. Evaluate $\int \frac{2x+5}{(x-1)(x+2)} dx$.

36. Find $B + C$ in $\frac{x^2}{(x-2)(x^2+1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1}$.

37. Decompose $\frac{1}{x(x^2+1)}$.

38. Decompose $\frac{x^3+x^2+1}{x(x^2+1)}$.

39. Find coefficient of x^n in expansion of $\frac{1}{(1-x)(1-2x)}$.

40. Decompose $\frac{4x}{x^4+4}$.

41. Find sum of coefficients in $\frac{7x^2-3x+5}{(x-1)(x-2)(x-3)}$.

42. Evaluate $\int \frac{dx}{x(x^4+1)}$.

43. Decompose $\frac{x^2}{(x^2+1)^2}$.

44. Find A in $\frac{x^3+1}{x^3-1} = 1 + \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$.

45. Decompose $\frac{1}{(x-1)(x^2+x+1)}$.

Level 4: EAMCET Speed Drills (15 Rapid Questions)**EAMCET SPEED**

46. If $\frac{x}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$, find $A + B$.

47. If $\frac{3x+1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$, find B .

48. If $\frac{1}{x(x+1)(x+2)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2}$, find B .

49. If $\frac{x^2+2}{(x-1)^3} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}$, find C .

50. If $\frac{x^2+5}{(x^2+1)(x^2+3)} = \frac{A}{x^2+1} + \frac{B}{x^2+3}$, find A .

51. If $\frac{x^3}{(x-1)(x-2)} = x + k + \frac{A}{x-1} + \frac{B}{x-2}$, find k .

52. If $\frac{1}{(x-1)(x-2)(x-3)(x-4)} = \sum \frac{A_k}{x-k}$, find $\sum A_k$.

53. If $\frac{2x+3}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$, find $A \cdot B$.

54. If $\frac{1}{x^2-16} = \frac{A}{x-4} + \frac{B}{x+4}$, find $A - B$.

55. If $\frac{x+2}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$, find A .

56. If $\frac{x^2+1}{(x-1)^2} = A + \frac{B}{x-1} + \frac{C}{(x-1)^2}$, find $A + B + C$.

57. If $\frac{1}{x(x^3+1)} = \frac{A}{x} + \frac{Bx^2}{x^3+1}$, find $A + B$.

58. If $\frac{5x-2}{x^2-4} = \frac{A}{x-2} + \frac{B}{x+2}$, find $A + B$.

59. If $\frac{1}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$, find C .

60. If $\frac{x^4+1}{x^2-1} = Q(x) + \frac{R(x)}{x^2-1}$, find $Q(2)$.

Level 5: JEE Advanced Multi-Concept Problems (15 Questions)**JEE ADVANCED**

61. Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$.

62. Find $\lim_{x \rightarrow \infty} x \sum_{k=1}^n \frac{k^2}{(x-k)}$.

63. Decompose $\frac{1}{x^4+x^2+1}$.

64. Find coefficient of x^{10} in $\frac{1}{(1-x)^3(1+x)}$.

65. Evaluate $\int_0^1 \frac{x^2}{(x^2+1)(x+1)} dx$.

66. If $P(x) = (x-1)(x-2) \dots (x-10)$, evaluate $\sum_{r=1}^{10} \frac{r^9}{P'(r)}$.

67. Decompose $\frac{x^4-1}{(x^2+1)^2(x^2+2)}$.

68. Evaluate $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{k^4+k^2+1}$.

69. Find the generating function representation of $\frac{1}{1-3x+2x^2}$.

70. Find A_1 in $\frac{1}{(x-1)^4(x+2)} = \sum_{k=1}^4 \frac{A_k}{(x-1)^k} + \frac{B}{x+2}$.

71. Decompose $\frac{x^2-1}{x^4+1}$ using symmetrical form.

72. Evaluate $\int \frac{dx}{x(x^2+1)}$.

73. Decompose $\frac{1}{(x-a)(x-b)(x-c)(x-d)}$.

74. Prove $\sum_{k=0}^n \frac{(-1)^k \binom{n}{k}}{x+k} = \frac{n!}{x(x+1) \dots (x+n)}$.

75. Evaluate $\sum_{r=1}^n \frac{r}{(r-1)!+r!+(r+1)!}$ using partial fractions.

 Master Answer Key & Quick Hints**Level 1:**

1. $\frac{1}{x-3} - \frac{1}{x-2}$
2. $\frac{-1/3}{x+1} + \frac{7/3}{x-2}$
3. $A = 4/3$
4. $B = 4/7$
5. $\frac{3/2}{x-1} - \frac{6}{x-2} + \frac{9/2}{x-3}$
6. $A + B = 4$
7. $\frac{1/6}{x-3} - \frac{1/6}{x+3}$
8. $\frac{1/2}{x-4} + \frac{1/2}{x+4}$
9. $A = 1/2$
10. $\frac{-3/2}{x+1} + \frac{7/2}{x+3}$
11. $C = 5$
12. $\frac{-4}{x-1} + \frac{11}{x-2}$
13. Sum = 3
14. $\frac{1}{x-1} - \frac{1}{x+3}$
15. $B = -4/3$

Level 2 & 3:

16. $\frac{1}{x-1} + \frac{3}{(x-1)^2}$
17. $C = 5$
18. $\frac{1}{x+1} + \frac{1}{(x+1)^2} - \frac{1}{(x+1)^3}$
19. $1 + \frac{1}{x-1} - \frac{1}{x+1}$
20. Quotient = x
31. $\frac{1/2}{x-1} - \frac{1/2(x-1)}{x^2+1}$
32. $A = 1/5$
33. $\frac{-1}{x^2+2} + \frac{2}{x^2+3}$
34. $A + B = 1$
39. $2^{n+1} - 1$
40. $\frac{1}{x^2-2x+2} - \frac{1}{x^2+2x+2}$
41. Sum = 7
42. $\frac{1}{4} \ln \left| \frac{x^4}{x^4+1} \right| + C$

Level 4 & 5:

46. $A + B = 1$
47. $B = 10$
48. $B = -1$
49. $C = 3$
50. $A = 2$
51. $k = 3$
52. $\sum A_k = 0$
53. $A \cdot B = -5/4$
54. $A - B = 1/4$
55. $A = 3/2$
61. $1/4$
62. $n(n+1)/2$
66. 1
68. $1/2$
72. $\frac{1}{7} \ln \left| \frac{x^7}{x^7+1} \right| + C$

SECTION 14 — SPEED-TRAINING CHALLENGES (40 QUESTIONS)

**SPEED
WORKSHOP**

Part A: 20 Flash Challenges (Target: ≤ 30 Seconds)

#1: $\ln \frac{3x+5}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$, find $A + B$.

Ans: 3 |  Limit at infinity = leading coeff ratio = $3/1 = 3$.

#2: $\ln \frac{1}{(x-3)(x-4)} = \frac{A}{x-3} + \frac{B}{x-4}$, find A .

Ans: -1 |  Cover-up at $x=3$: $1/(3-4) = -1$.

#3: $\ln \frac{x}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2}$, find B .

Ans: 2 |  Numerator at $x=2$ is 2.

#4: $\ln \frac{x^2+1}{(x-1)(x-2)(x-3)} = \sum \frac{A_k}{x-k}$, find $\sum A_k$.

Ans: 1 |  Limit at infinity = 1.

#5: $\ln \frac{1}{x(x+1)(x+2)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2}$, find B .

Ans: -1 |  Binomial middle sign = -1.

#6: $\ln \frac{x^2+4}{(x^2+1)(x^2+3)} = \frac{A}{x^2+1} + \frac{B}{x^2+3}$, find $A + B$.

Ans: 1 |  Set $u = x^2$, limit $u \rightarrow \infty$ is 1.

#7: $\ln \frac{2x^3}{x^2-1} = Q(x) + \frac{R(x)}{x^2-1}$, find $Q(x)$.

Ans: 2x |  Leading term $2x^3 / x^2 = 2x$.

#8: $\ln \frac{1}{(x-1)^3(x+2)} = \sum \frac{A_k}{(x-1)^k} + \frac{B}{x+2}$, find B .

Ans: -1/27 |  Cover-up at $x=-2$: $1/(-3)^3 = -1/27$.

#9: $\ln \frac{5x-2}{x^2-4} = \frac{A}{x-2} + \frac{B}{x+2}$, find A .

Ans: 2 |  Cover-up at $x=2$: $(10-2)/4 = 2$.

#10: $\ln \frac{1}{x^2-25} = \frac{A}{x-5} + \frac{B}{x+5}$, find $A - B$.

Ans: 1/5 |  $A = 1/10$, $B = -1/10 \rightarrow A - B = 2/10 = 1/5$.

#11: $\ln \frac{x^2}{(x-1)^3} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}$, find C .

Ans: 1 |  Evaluate x^2 at $x=1 \rightarrow 1$.

#12: $\ln \frac{3}{(x-1)(x-2)(x-3)} = \sum \frac{A_k}{x-k}$, find $\sum A_k$.

Ans: 0 |  $\deg(P) < \deg(Q) - 1 \rightarrow \text{sum} = 0$.

#13: $\ln \frac{x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$, find A .

Ans: 1 |  Cover-up at $x=0 \rightarrow 1/1 = 1$.

#14: $\ln \frac{4x}{x^2-1} = \frac{A}{x-1} + \frac{B}{x+1}$, find A .

Ans: 2 |  Cover-up at $x=1 \rightarrow 4/2 = 2$.

#15: $\ln \frac{x^2+2}{(x+1)^2} = A + \frac{B}{x+1} + \frac{C}{(x+1)^2}$, find A .

Ans: 1 |  Ratio of leading coefficients = 1.

#16: $\ln \frac{1}{(x-a)^2(x-b)} = \dots + \frac{C}{x-b}$, find C .

Ans: 1/(a-b)^2 |  Cover-up at $x=b \rightarrow 1/(b-a)^2$.

#17: $\ln \frac{2x+3}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$, find B .

Ans: 1/3 |  Cover-up at $x=-2$: $(-4+3)/(-3) = 1/3$.

#18: $\ln \frac{x^4}{x^4-1} = 1 + \dots$, find the constant term of the quotient.

Ans: 1 |  Term matching $1 + 1/(x^4-1)$.

#19: $\ln \frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$, find $A \cdot B$.

Ans: -1 |  $A=1$, $B=-1 \rightarrow \text{product} = -1$.

#20: $\ln \frac{3x^2+1}{(x-1)(x-2)(x-4)} = \sum \frac{A_k}{x-k}$, find $\sum A_k$.

Ans: 3 |  Leading coeff ratio = 3.

⚡ Part B: 20 Speed Drills (Target: ≤ 60\$ Seconds)

#21: Decompose $\frac{x^2+3x+5}{(x-1)^3}$.

Ans: $\frac{1}{x-1} + \frac{5}{(x-1)^2} + \frac{9}{(x-1)^3}$

💡 **Mental Step:** Put $x = 1+y \rightarrow (1+y)^2 + 3(1+y) + 5 = y^2 + 5y + 9$.

#23: Find A, B in $\frac{x^2+2}{(x^2+1)(x^2+4)} = \frac{A}{x^2+1} + \frac{B}{x^2+4}$.

Ans: **A = 1/3, B = 2/3**

💡 **Mental Step:** Set $u = x^2$, Cover-up at $u=-1$ and $u=-4$.

#25: Decompose $\frac{x^3+1}{x^2-1}$.

Ans: $x + \frac{1}{x-1}$

💡 **Mental Step:** Factor $x^3+1 = (x+1)(x^2-x+1)$. Cancel $(x+1) \rightarrow (x^2-x+1)/(x-1) = x + 1/(x-1)$.

#27: Find A_1 in $\frac{2x+1}{(x-1)^2(x+2)}$.

Ans: **1/9**

💡 **Mental Step:** Derivative at $x=1$ of $(2x+1)/(x+2)$.

#29: Find A in $\frac{1}{x^4-1} = \frac{A}{x-1} + \dots$

Ans: **1/4**

💡 **Mental Step:** General Heaviside: $1/Q'(1) = 1/(4(1)^3) = 1/4$.

#31: Decompose $\frac{4x}{x^4+4}$.

Ans: $\frac{1}{x^2-2x+2} - \frac{1}{x^2+2x+2}$

💡 **Mental Step:** Difference of Sophie Germain factors.

#33: Find $A + B$ in $\frac{x}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$.

Ans: **1**

💡 **Mental Step:** Must-Memorize Identity 2 sum = 1.

#35: Decompose $\frac{1}{x(x^2+4)}$.

Ans: $\frac{1/4}{x} - \frac{1/4x}{x^2+4}$

💡 **Mental Step:** Must-Memorize Identity 9 with $a=2$.

#37: Sum $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$.

Ans: **1/4**

💡 **Mental Step:** $1/(2*1*2) = 1/4$.

#39: Find $A - B$ in $\frac{1}{(x-1)(x-3)}$.

Ans: **-1**

💡 **Mental Step:** $A = -1/2, B = 1/2 \rightarrow A-B = -1$.

#22: Evaluate $\int \frac{dx}{(x-1)(x-3)}$.

Ans: $\frac{1}{2} \ln \left| \frac{x-3}{x-1} \right| + C$

💡 **Mental Step:** Must-Memorize Identity 1 with difference = 2.

#24: Find $Bx + C$ in $\frac{x}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$.

Ans: $-\frac{1}{2}x + \frac{1}{2}$

💡 **Mental Step:** $A = 1/2$. Put $x = i \rightarrow i/(i-1) = Bi+C \rightarrow C=1/2, B=-1/2$.

#26: Sum $\sum_{r=1}^n \frac{1}{r(r+1)}$.

Ans: $\frac{n}{n+1}$

💡 **Mental Step:** Telescoping: $1 - 1/(n+1) = n/(n+1)$.

#28: Evaluate coefficient of x^n in $\frac{1}{(1-x)(1-2x)}$.

Ans: **$2^{n+1} - 1$**

💡 **Mental Step:** $2/(1-2x) - 1/(1-x) \rightarrow 2(2^n) - 1$.

#30: Find $A + B + C$ in $\frac{x^2-5x+2}{(x-1)(x-2)(x-3)}$.

Ans: **1**

💡 **Mental Step:** Leading coefficient ratio = 1.

#32: Evaluate $\int \frac{dx}{x(x^5+1)}$.

Ans: $\frac{1}{5} \ln \left| \frac{x^5}{x^5+1} \right| + C$

💡 **Mental Step:** Standard x^n identity with $n=5$.

#34: Find A in $\frac{x^3+5}{x^3-32}$ under $(x-2)$.

Ans: **13/80**

💡 **Mental Step:** $P(2)/Q'(2) = 13 / (5*16) = 13/80$.

#36: Find C in $\frac{5x^2}{(x-2)^3}$.

Ans: **20**

💡 **Mental Step:** $P(2) = 5(4) = 20$.

#38: Decompose $\frac{x^2-1}{x^4+1}$.

Ans: $\frac{1}{2\sqrt{2}} \left[\frac{x}{x^2-\sqrt{2}x+1} - \frac{x}{x^2+\sqrt{2}x+1} \right]$

💡 **Mental Step:** Divide by x^2 , set $t = x+1/x$.

#40: Find $A + B + C + D$ in $\frac{x^3+2x}{(x^2+1)^2}$.

Ans: **2**

💡 **Mental Step:** Put $t = x^2+1 \rightarrow x/t + x/t^2 \rightarrow A=1, B=0, C=1, D=0 \rightarrow \text{Sum} = 2$.

SECTION 15 — PREVIOUS-YEAR PATTERN ANALYSIS & EXAM STRATEGY

TREND FORENSICS

 Frequency Breakdown by Denominator Structure

Based on analysis of over 15+ years of JEE Main, JEE Advanced, and TS/AP EAMCET examinations:

Denominator Structure	EAMCET %	JEE Main %	JEE Advanced %	Key Tested Skill
Distinct Linear $(x - a)(x - b)$	45%	20%	10%	Speed Cover-Up, Option Elimination
Even Powers $(x^2 + a)(x^2 + b)$	25%	35%	15%	Algebraic $u = x^2$ Substitution
Repeated Linear $(x - a)^n$	15%	25%	25%	Taylor Shift $x = a + y$, Derivatives
Telescoping Series $\frac{1}{r(r+1)\dots}$	5%	15%	30%	Series Summation, Difference Method
Symmetric / Factorization $(x^4 + 4)$	10%	5%	20%	Sophie Germain, Palindromic $t = x + 1/x$

 High-Priority Topics

- Heaviside Cover-Up Rule on linear factors.
- Infinity limits for coefficient sums $\sum A_i$.
- Even power $u = x^2$ substitution.
- Standard integral identities $\int \frac{dx}{x(x^n+1)}$.
- Taylor shift for repeated roots $(x - a)^n$.
- Telescoping partial fractions in sequences.

 Low-Priority Topics (Do Not Over-invest Time)

- Large 4x4 or 5x5 undetermined coefficient matrices.
- Factoring non-standard 5th degree polynomials without rational roots.
- Long polynomial division when synthetic / term-matching suffices.
- Deriving standard identities from scratch in exams.

SECTION 16 — MASTER CHEAT SHEET & 7-DAY MASTERY PLAN

ULTIMATE COMMAND CENTER

 **PARTIAL FRACTIONS — COMPLETE SHORTCUT MASTER SHEET**

1. Standard Linear Cover-Up:
$$A_k = \frac{P(x)}{Q(x) \text{ without } (x-a_k)} \Big|_{x=a_k} = \frac{P(a_k)}{Q'(a_k)}$$

2. Repeated Root Taylor Shift:

For $\frac{P(x)}{(x-a)^n}$, put $x = a + y$, expand & divide by y^n .

3. Even Symmetry Shortcut:

For all even powers, substitute $u = x^2$ \$\implies\$ decompose linear in u .

4. Infinity Limit for Coefficient Sums:
$$\sum_{i=1}^n A_i = \lim_{x \rightarrow \infty} x \cdot f(x) = \frac{\text{Leading coeff of } P}{\text{Leading coeff of } Q}$$

5. Complex Roots on Quadratics:

For $(x^2 + a^2)$, plug $x = ia$ and equate real & imaginary parts.

6. Calculus Power Rule:
$$\int \frac{dx}{x(x^n+1)} = \frac{1}{n} \ln \left| \frac{x^n}{x^n+1} \right| + C$$

 **7-Day Partial Fractions Mastery Schedule**

Follow this high-intensity daily schedule to achieve 100% accuracy and 30-second speed:

Day	Focus Area	Target Tasks & Practice	Success Milestone
Day 1	Concepts & Templates	Study Section 1 & 2. Master Proper/Improper & Degree rules.	0 template errors
Day 2	Basic Decompositions	Solve Progressive Practice Level 1 & 2 (30 Questions).	< 40s per linear problem
Day 3	The 20 Shortcuts	Master Section 3 & 4. Memorize the Top 10 Identities.	Instant Cover-Up recall
Day 4	JEE Main & EAMCET Speed	Solve Section 6 (25 JEE Main) + Section 8 (25 EAMCET).	< 30s for EAMCET MCQs
Day 5	JEE Advanced Deep Dive	Study Section 5 & Solve Section 7 (20 JEE Advanced).	Master Taylor Shift & Residues
Day 6	Mixed & Speed Drills	Solve Section 14 (40 Speed Challenges) & Traps review.	Zero common traps triggered
Day 7	Mastery & Mock Test	Revise Section 11 & 16. Complete 30-minute timed mock test.	100% speed & accuracy score

Competitive Mathematics Faculty Handbook

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